

Social Factors of Health Covary with Population Stratification and Confound Heritability Estimates

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Abstract

Social factors of health (SFOH) are critical determinants of human traits but are rarely incorporated into genetic models. Here, we assess how participant-completed SFOH survey data influence heritability estimates and trait variance in 85,963 diverse All of Us participants spanning multiple ancestries and racial identities. We summarized SFOH survey data using multiple correspondence analysis (MCA) into axes representing distinct social dimensions (e.g., social support, perceived stress) and developed a Social Similarity Index (SSI) to capture social environment similarity among individuals. To analyze trait heritability, we used Haseman-Elston (HE) regression for its sensitivity to residual structure. Including SFOH measures as covariates in HE regression models significantly reduced heritability estimates for four of 18 traits, three of which are anthropometric: body mass index, hip circumference, waist circumference, and HDL cholesterol. This suggests that unmodeled social structure can be misattributed to genetic effects. Though several SFOH measures have non-zero heritability when adjusting for three genetic principal components ($h^2 \approx 0.01-0.09$), these are reduced to similar estimates when adjusting for seven genetic principal components ($h^2 \approx 0.01-0.02$). This convergence across SFOH measures, which capture different social dimensions, indicates that SFOH measures capture population stratification, emphasizing their utility for reducing confounding in heritability estimates. SFOH measures were also associated with trait variance, with anthropometric traits—those with reduced heritability—exhibiting the most extensive dispersion effects. This implies that social environment influences trait variability. Our findings highlight the necessity of including social and environmental factors in genetic studies to reduce potential confounding.

39 Significance Statement

40 Modeling the social drivers of human traits remains a critical challenge in
41 biomedicine. This is particularly acute in statistical genetics, where genome-wide
42 studies routinely adjust for genetic stratification while social and environmental
43 exposures remain unmeasured or oversimplified. Using 85,963 All of Us participants,
44 we show that social factors of health significantly reduce trait heritability estimates,
45 demonstrating that unmodeled social structure can inflate genetic heritability esti-
46 mates. Social factor measures themselves appear to be heritable before accounting
47 for population structure, revealing that their genetics primarily capture population
48 stratification. These findings highlight that robust genetic research requires mea-
49 suring and adjusting for social environment factors to avoid misattributing social
50 influences as genetic effects.

51 1 Introduction

52 Modeling the social drivers of human traits remains a critical challenge in
53 biomedicine. This is particularly acute in statistical genetics, where genome-wide
54 studies routinely adjust for genetic stratification [1–4], while trait-relevant social
55 and environmental exposures typically remain unmeasured or oversimplified. Social
56 factors of health (SFOH)—such as housing, financial insecurity, education, and per-
57 ceived stress—shape both disease risk and access to health care [5–9]. SFOH are
58 increasingly captured in structured survey form in biobank-scale cohorts, includ-
59 ing the All of Us Research Program [10]. However, SFOH data are rarely explicitly
60 incorporated in heritability and genetic association models [11, 12]. Like genetic
61 variation, these factors are stratified within populations but can change rapidly and
62 profoundly impact health outcomes. This omission can leave residual trait varia-
63 tion structured by social factors, such that social stratification is misattributed as
64 genetic variance, creating incomplete inferences about underlying genetic architec-
65 ture shaped by spurious signals. Although the All of Us Social Determinants of
66 Health (SDOH) survey data has been previously characterized [13] and connected
67 to health outcomes [14, 15], to our knowledge, it has not been incorporated in
68 heritability models.

69 Survey data often contain trait-relevant information, but their relative struc-
70 ture and information content differ from those of genetic [16] data and from the
71 assumptions of standard genetic models, making joint modeling challenging. A
72 recent work applied PCA to dietary survey data and used PC1 as an environmental
73 factor in genetic association testing [17]. Dual likelihood modeling frameworks [18]
74 can address some of these technological challenges. Furthermore, survey data have
75 known participation bias [19] as well as completion and non-response rates that
76 are correlated with demographic factors, such as race and ethnicity [20]. Because
77 survey completion rates vary by race and ethnicity, it becomes difficult to deter-
78 mine whether observed patterns reflect true population-specific factors or simply
79 artifacts of differential survey participation. This challenge is particularly acute in
80 multi-ancestry studies. Related methodological work has addressed the need for
81 ancestry-aware approaches when modeling complex traits [21].

82 Stratifying analyses by self-identified race or ethnicity can itself be problematic,
83 as these social constructs overlap substantially with genetic ancestry patterns yet do
84 not map cleanly onto them, creating challenges in accurately attributing variance

85 components [22–24]. Recent methodological advances have developed extensions
 86 of Haseman-Elston regression for estimating heritability to account for population
 87 stratification while assuming constant heritability across subpopulations [25]. How-
 88 ever, this assumption might not hold when social and genetic stratification are
 89 entangled, as we investigate here. Rather than attempting to disentangle these over-
 90 lapping signals through stratification, we conduct our analysis in the multi-ancestry
 91 and multi-racial group of 85,963 participants, allowing us to characterize how SFOH
 92 and genetic factors jointly structure trait variation across the social and genetic
 93 diversity of these individuals. SFOH survey data contains information that is poten-
 94 tially correlated with genetic data, and understanding the relationship between the
 95 two is critical for robust joint analysis while avoiding additional bias in genetic
 96 estimates.

97 Here, we investigate how high-dimensional social structure captured by the All
 98 of Us SDOH survey influences single nucleotide polymorphism-based heritability
 99 (h_g^2)—the proportion of phenotypic variance in a population explained by additive
 100 genetic effects of common SNPs—of 18 anthropometric and metabolic traits. We
 101 use the terms “heritability” and h^2 to refer to this SNP-based heritability (h_g^2),
 102 which provides an estimate for narrow-sense heritability. In this paper, we address
 103 three specific questions: (1) Do SFOH measures themselves have non-zero heritabil-
 104 ity, suggesting potential confounding between social and genetic structures? (2)
 105 Does accounting for SFOH reduce trait heritability estimates, indicating that social
 106 structure may be misattributed to genetic effects? (3) Do SFOH measures influence
 107 trait variance patterns beyond their effects on means? To address these questions,
 108 we summarize the SDOH survey responses into SFOH measures using two comple-
 109 mentary strategies: dimensionality reduction via multiple correspondence analysis
 110 (MCA) and a Social Similarity Index (SSI) derived from an environmental relat-
 111 edness matrix (ERM). These yield participant-level covariates that integrate the
 112 critical yet often overlooked influences of social environment factors and facilitate
 113 their joint analysis with genetic factors.

114 2 Results

115 We analyzed data from 85,963 All of Us (v7) participants who met the following
 116 criteria: availability of SDOH survey data, genotyping array data, a selected sex at
 117 birth of “female” or “male” (Table S1), and a self-identified race (Table S2) from The
 118 Basics survey. We performed analyses of collinearity and the relationship between
 119 social and population genetic structure on this combined analysis group, while her-
 120 itability estimation was restricted to subsets of individuals with measurements for
 121 each of the 18 analyzed quantitative traits.

122 2.1 Collinearity

123 To characterize the relationship between social and genetic measures, we first
 124 derived two sets of SFOH measures from one-hot encoded All of Us SDOH survey
 125 data: the SFOH SSI and SFOH MC axes. We then assessed collinearity by calcu-
 126 lating Spearman correlations between these SFOH MC axes and genetic principal
 127 components (PCs) (Fig. 1). For the SSI, the maximum absolute Pearson correla-
 128 tion with any genetic PC was 0.317 and all variance inflation factors (VIFs), where
 129 values above 5 signal moderate multicollinearity, remained below 1.150, indicating

an acceptable amount of multicollinearity. For the top forty SFOH MC axes, the maximum correlation with any genetic PC was 0.011 (VIFs < 1.145). Thus, each SFOH measure can safely be modeled alongside genetic PCs without problematic multicollinearity.

2.2 Missingness Analysis

We excluded missing responses coded as “PMI: Skip”, “PMI: Don’t Know”, “PMI: Not Sure”, “PMI: Other” or “PMI: Prefer Not To Answer”. At the question-level, missingness was very low (less than 5%) for nearly all themed items (Table S3), with only two un-themed questions exhibiting high skip rates. Theme-level missingness was similar across self-identified race groups (2–8%) (Table S4), and at the individual level the median per-participant missingness was approximately 3%, with less than 1% of participants missing more than 20% of items.

2.3 Trait heritability

Using GCTA [26], we estimate heritability using Haseman–Elston (HE) regression [27], a computationally efficient method of moments estimator suitable for large datasets. We chose HE regression over the genomic-relatedness-based restricted maximum-likelihood (GREML) approach because its sensitivity to residual structure makes it well-suited for detecting when SFOH measures capture variance that would otherwise be misattributed to genetic effects—our primary research question.

To assess how SFOH measures and genetic PCs influence these estimates, we tested six nested covariate models (Table 1) that progress from basic demographic adjustment—age, sex, self-identified race, and self-identified ethnicity (model 1)—to full covariate sets (model 6). All analyses of systolic blood pressure include adjustment for medication use. The single-statistic SFOH SSI only appears in two models (2, 5) while the partitioned multi-axis SFOH MCA appears in four models (2, 3, 5, and 6). The lead SFOH MC axes 1:6 (14.73% variance explained) (Fig. S1), appear in models 2 and 5, while the comprehensive SFOH MC axes 1:40 (38.02% variance explained), appear in models 3 and 6. The top genetic PCs 1:7 (96.84% of variance explained) (Fig. S2) are used in models 4, 5, and 6.

SFOH Adjustment Model	SFOH Covariates
1. No Adjustment	None
2. Lead Measures	SFOH MC axes 1:6 or SFOH SSI
3. Comprehensive Measures	SFOH MC axes 1:40
4. No Adjustment + Genetic PCs 1:7	None
5. Lead Measures + Genetic PCs 1:7	SFOH MC axes 1:6, or SFOH SSI
6. Comprehensive Measures + Genetic PCs 1:7	SFOH MC axes 1:40

Table 1: Covariate models used in trait heritability analyses. Models build progressively from no SFOH adjustment to full SFOH adjustment.

2.3.1 HE regression is sensitive to unmodeled SFOH

We used HE regression to estimate heritability via cross product (HE-CP). In the combined analysis group of up to 85,963 individuals (Fig. 2), adding genetic PCs 1:7 (models 4, 5, and 6) consistently reduced heritability estimates relative to the

Base (model 1) and SFOH-only models (models 2 and 3). These reductions were significant in 11 of 18 traits: BMI, HDL cholesterol, carbon dioxide, creatinine, diastolic blood pressure, glucose, hip circumference, heart rate, sodium, urea nitrogen, and waist circumference. Sex-stratified analyses revealed similar patterns, with minor variation in estimates between females (Fig. S3) and males (Fig. S4) for certain traits. This pattern indicates that genetic PCs capture population structure that inflates heritability estimates when unaccounted for, which is a known property of HE regression. When genetic ancestry is associated with trait values, this method may misattribute this covariance to genetic factors; including genetic PCs as covariates corrects for this potential source of confounding.

Adding SFOH MC axes to the Base model (models 2 and 3) also reduced the heritability estimate of 5 of 18 traits—BMI, hip circumference, waist circumference, glucose, and HDL cholesterol (Fig. 3)—while the SSI had no significant effect on any trait (Table S5). This suggests that a global measure of social similarity is less consequential for heritability estimation than specific social dimensions. Specifically, including SFOH MC axes 1:6 (model 2) significantly reduced heritability estimates relative to the Base model (model 1) for BMI (by 0.113 from 0.268 to 0.155), hip circumference (by 0.093 from 0.302 to 0.209), and waist circumference (by 0.107 from 0.339 to 0.232). Including the MC axes 7:40 (model 3) further reduced the heritability of these traits, though not significantly (Table S6). As a sensitivity analysis, we also performed PCA on the survey data [17], here we use the same one-hot encoded SFOH survey data. When including SFOH PCs into heritability models, results were broadly consistent with MCA findings (Fig. S5).

The sensitivity of HE regression to SFOH adjustment is consistent with SFOH measures capturing social structure that correlates with genetic similarity. Our finding that specific SFOH measures initially appear heritable but decrease with comprehensive genetic PC adjustment (Section 2.4) indicates this correlation primarily reflects population stratification rather than true genetic effects on social factors. Because HE regression attributes any phenotypic covariance aligned with the GRM to additive genetic variance, SFOH measures that proxy genetic stratification can substantially influence estimates when included as covariates. Including SFOH measures removes this shared variance, leading to lower h^2 estimates even without genetic PCs. This demonstrates that SFOH measures can act as proxies for confounding population structure that would otherwise be misattributed to genetic effects.

These analyses demonstrate that survey-derived SFOH data summarized with MCA can influence heritability estimates obtained via HE regression, sometimes significantly so. Our results suggest that genetic PCs and SFOH measures capture overlapping information about population stratification, with SFOH measures potentially providing additional adjustment for residual structure not fully captured by a limited number of genetic PCs. We observed a similar pattern in an ancestry-specific analysis restricted to 48,146 individuals of European ancestry based on their genetic similarity to European reference populations in gnomAD, HGDP, and 1000 Genomes who additionally identified as “White” and “Not Hispanic or Latino”, which we refer to as the “European-ancestry subgroup”. For this analysis, the SFOH MC axes were recalculated using only the survey data from this subgroup of individuals. Despite reduced statistical power and larger standard errors, the pattern seen

in the combined group analysis persisted (Fig. S6), suggesting that the confounding effect of social structure is not an artifact of multi-ancestry analysis alone.

2.4 SFOH measures show progressive reduction in heritability with genetic PC adjustment

To assess whether the SFOH measures are estimated to have non-zero heritability, we applied HE regression the SFOH MC axes and the SSI as traits. We tested three nested covariate models (Table 2) to assess how basic demographics and genetic PCs influence these estimates.

Model	Covariates
1.	Base: Age + Sex + Race + Ethnicity
2.	Base + Genetic PCs 1:3
3.	Base + Genetic PCs 1:7

Table 2: Covariate models used in SFOH measure heritability analyses.

2.4.1 SFOH heritability estimates decrease with comprehensive genetic PC adjustment

Initial heritability estimates for many SFOH measures were significantly different from zero when adjusting only for basic demographics (Fig. S7 and Fig. S8), some substantially so with estimates exceeding $h^2 = 0.600$. However, these estimates decreased dramatically upon inclusion of genetic PCs. For MC5, heritability decreased substantially (from $h^2 = 0.086$ to 0.013) when increasing adjustment from three to seven genetic PCs. Notably, with seven PCs, heritability estimates for all major SFOH measures converged to a similarly low baseline (Fig. 4, Table S7). We observed a similar pattern in the European-ancestry subgroup using the subgroup-specific MC axes (Fig. S9), though with reduced statistical power, reinforcing that the non-zero heritability of SFOH MCs primarily reflects population stratification, which operates both between and within ancestral groups. The latent structure captured by MCA showed strong consistency between the multi-ancestry combined analysis group MC axes and the European-ancestry subgroup MC axes (Fig. S10), with corresponding dimensions exhibiting moderate to high correlations.

This progressive reduction and convergence to similar estimates across measures (Table S8) capturing distinct social dimensions indicate that the apparent genetic signal in SFOH measures primarily reflects population stratification rather than trait-specific genetic architectures. If these measures captured genuine genetic effects on different social factors—neighborhood environment, loneliness, stress, discrimination—we would expect their heritability estimates to diverge after controlling for global ancestry structure. Instead, their convergence to a similar baseline suggests a single underlying confounder. The small residual heritability observed even with seven genetic PCs most likely represents incomplete control for fine-scale population structure, though genuine but minimal genetic effects on social factors or technical artifacts from survey data structure cannot be ruled out. This pattern

245 of decrease and convergence was recapitulated when using PCA to summarize one-
246 hot encoded SFOH data (Fig. S11), confirming that it is not an artifact of MCA
247 but a robust property of the social data structure.

248 **2.4.2 The SSI demonstrates convergence without affecting trait** 249 **heritability**

250 The SSI demonstrates an important pattern. While it shows apparent heritability
251 that, like the SFOH MC axes, decreases with genetic PC adjustment, it does not
252 reduce trait heritability estimates when included as a covariate (Fig. S12). This
253 distinction is informative. The SSI captures overall social similarity across all survey
254 dimensions, and its correlation with genetic PCs is comparable to individual MC
255 axes. Yet unlike specific MC axes, it does not affect trait heritability. This pattern
256 suggests that correlation with population structure is necessary but not sufficient
257 for SFOH measures to influence trait heritability estimates. Rather, measures must
258 capture specific social dimensions that structure trait variation to act as effective
259 covariates. The SSI result therefore supports the interpretation that reductions in
260 trait heritability reflect reduction of genuine social confounding aligned with genetic
261 structure rather than an artifact of including ancestry-correlated covariates.

262 **2.5 SFOH measures and fundamental demographics strongly** 263 **influence trait variance**

264 We applied double generalized linear models to investigate how SFOH measures and
265 demographics influence the dispersion of anthropometric and metabolic traits. All
266 models included comprehensive adjustment for age, sex, race, and ethnicity while
267 testing the two distinct SFOH measures: 20 SFOH MCs, and the SSI.

268 **2.5.1 Comprehensive dispersion effects across SFOH measures**

269 All SFOH measures showed widespread significant effects on trait dispersion after
270 multiple testing correction. Multiple MC axes showed widespread significant effects
271 on trait dispersion (Fig. 5). In particular, SFOH MC axes demonstrate strong dis-
272 persion effects on anthropometric measures and cardiovascular traits. MCs 1 and 2
273 show broad significance, particularly on metabolic traits. Many MC axes reached
274 nominal significance and a substantial amount survived stringent test correction.

275 The SSI also demonstrated significant dispersion effects on traits (Fig. S13).
276 Anthropometric measures, particularly hip and waist circumference show the
277 strongest and most consistent association with the SSI. The SSI consistently reached
278 overall significance after stringent test correction.

279 The two SFOH measurements: MC axes and SSI suggest that SFOH influence
280 trait variance through both multi-dimensional approaches and a broader single sum-
281 mary. The convergence of both measures on anthropometric traits reinforces the
282 robust connection between SFOH and body composition variance independent of
283 demographic factors.

284 **2.5.2 Comprehensive dispersion effects of demographics**

285 The demographic covariates we considered—age, sex, race, and ethnicity—
286 contributed significantly to trait dispersion, independent of SFOH measures. Age
287 demonstrated the most widespread influence across traits, suggesting that vari-
288 ance patterns systematically change with age. Sex, race, and ethnicity showed more

targeted, but still significant, dispersion effect, indicating that demographic factors contribute to trait variance beyond their associations with SFOH measures. Further, demographic factors overall showed a greater magnitude of effect sizes, indicating that they are highly influential in trait variance provided they remain significant. The persistence of significant demographic factors after comprehensive SFOH adjustment underscores the importance of including these covariates to isolate true SFOH measure influences. Furthermore, the pattern of positive and negative effects across demographic covariates suggests that different groups may have distinct contributions to trait variance, highlighting the multidimensional nature of how demographics and SFOH shape trait variance.

2.6 Assessing the association between SFOH and genetic population structure

We assessed the extent to which SFOH and genetic population structure are associated by examining bidirectional relationships between SFOH MCs and genetic PCs in the combined analysis group.

2.6.1 SFOH measures and genetic PCs show bidirectional but limited associations in the combined analysis group

First, to assess how social factors explain genetic structure, we regressed each of genetic PCs 1:10 on SFOH derived MC axes 1:40. A linear combination of MC axes explained non-trivial variation in genetic PC1 ($R^2 = 0.206$), likely reflecting alignment between global ancestry and social groupings (Table S9). The variance explained by MC axes for genetic PCs 2:10 was considerably lower (max R^2 for any genetic PC is 0.043).

Next, to assess how genetic structure explains social factor patterns, we regressed each of MC axes 1:20 on genetic PCs 1:10. Genetic PCs explained varying amounts of variation in MC structure. All twenty MCs showed statistically significant associations with genetic PCs, with R^2 values ranging from 0.001 (MC12) to 0.076 (MC3) (Table S10). Several MCs demonstrated notable strong genetic associations, including the aforementioned MC3, MC4 ($R^2 = 0.047$), MC1 ($R^2 = 0.044$), and MC16 ($R^2 = 0.035$). The consistent statistical significance across all components suggests broad genetic influence on the representation of SFOH measures. These axes likely capture SFOH topics such as feelings of healthcare and everyday discrimination, perceived stress, and neighborhood environment (Fig. 6).

2.6.2 Genetic PCs and SFOH measures constitute largely distinct sources of variation

These findings suggest that while SFOH MC axes and genetic PCs are statistically associated—particularly where social and ancestral structures align—the practical overlap is limited. Most of the variation in each domain does not covary with the other. This underscores the importance of treating social and genetic factors as distinct sources of variation in analyses of complex traits and reinforces the need to measure environmental exposures directly, rather than relying on race as a proxy. Further, this apparent distinction between domains supports the use of both SFOH MC axes and genetic PCs in downstream analyses.

332 **3 Methods**

333 **3.1 Data source and Cohort Selection**

334 We analyzed data from All of Us v7. To characterize the genetic ancestry structure
335 of this diverse cohort, we performed PCA and present the population centroids for
336 PC1 and PC2 (Fig. S14). For a detailed visualization of the genetic PC structure
337 of the All of Us dataset, we refer readers to Figures 1 and 2 in [28]. The dataset
338 includes over one billion genetic variants and has facilitated the identification of
339 genetic associations with various diseases.

340 We included individuals who had genetic data available, completed the Social
341 Determinants of Health survey, self-identified male or female sex at birth, and who
342 had reported a self-identified racial identity from The Basics survey. To maximize
343 sample size, we included individuals identifying as “Hispanic or Latino”. The final
344 combined analysis group comprised 85,963 individuals.

345 As a sensitivity analysis, we repeated key analyses in a subset of 48,146 individu-
346 als of European ancestry who additionally identified as “White” and “Not Hispanic
347 or Latino” which we refer to as the “European-ancestry subgroup”. We deter-
348 mined genetic ancestry using the All of Us genetic similarity categories from the
349 auxiliary file `ancestry_preds.tsv`, which group individuals based on their genetic
350 similarity to reference populations from the Genome Aggregation Database (gno-
351 mAD), Human Genome Diversity Project (HGDP), and 1000 Genomes. Specifically,
352 we used the “1KGP-HGDP-EUR-like (EUR or European) category to define the
353 European-ancestry subgroup.

354 **3.2 Quality Control**

355 Genotyping was performed using a custom Illumina array built on the design frame-
356 work of the Global Screening Array. The genome build used is hg38/GRCh38. Please
357 refer to the All of Us Research Program technical documentation and data release
358 notes for further details. We conducted variant filtering using PLINK 2.0 [29] on the
359 All of Us version 7 controlled-tier array data, restricted to autosomal SNPs. This
360 reassigned variant identifiers and removed all duplicate variants. To mitigate the
361 effects of linkage disequilibrium, we pruned variants using a sliding window approach
362 (200 variants per window, with a step size of 50 variants) and an r^2 threshold of 0.2.
363 We excluded variants with a minor allele frequency below 0.01, a genotype missing-
364 ness rate exceeding 1%, deviating from Hardy-Weinberg equilibrium ($p < 1 \times 10^{-6}$),
365 and individuals with more than 2% missing genotype calls. We assessed relatedness
366 up to the third degree using KING [30] (kinship coefficient ≥ 0.0625), and removed
367 one member from each related pair. After quality control, we retained 285,123 unre-
368 lated individuals and 56,961 high-quality common variants for genetic PC analysis
369 (Fig. S2).

370 **3.3 Trait Processing**

371 We selected 18 clinically relevant anthropometric and metabolic traits known to
372 demonstrate variation across self-identified racial groups, enabling us to investi-
373 gate how additional social factors might contribute to these patterns. These traits
374 include: alanine aminotransferase (AAT), alkaline phosphatase (AlkPhos), aspar-
375 tate aminotransferase (AspAminTrans), body mass index (BMI), carbon dioxide
376 (CO2), chloride (Cl), cholesterol in HDL (CholHDL), creatinine (Creat), diastolic

377 blood pressure (DBP), glucose, heart rate (HR), hematocrit, hip circumference (Hip-
 378 Circ), protein, sodium, systolic blood pressure (SBP), urea nitrogen (UreaNitro),
 379 and waist circumference (WC).

380 Following trait selection, we performed data cleaning and deduplication proce-
 381 dures prior to downstream analyses. For individuals with multiple measurements,
 382 we retained the most recent value. To mitigate the influence of extreme values, we
 383 calculated the IQR for each biomarker independently, and observations falling out-
 384 side these bounds were truncated to the corresponding threshold values. Systolic
 385 blood pressure was adjusted for antihypertensive medication using a binary indica-
 386 tor variable. Sample sizes varied by trait due to differential measurement availability
 387 across participants.

388 3.4 Estimating Variance Contribution by Genome-wide Complex 389 Trait Analysis (GCTA)

390 We estimated h_g^2 using HE regression as implemented in the Genome-wide Complex
 391 Trait Analysis software (GCTA). For our primary inference, we report HE-CP point
 392 estimates from HE regression and standard errors (SE) calculated by jackknifing
 393 over individuals. 95% confidence intervals (CIs) were constructed as $HE - CP \pm$
 394 $1.96(SE)$. To evaluate whether SFOH measures alter heritability estimates, and
 395 whether SFOH measures themselves may be heritable, we compared the 95% CIs
 396 between pairs of nested models. Non-overlapping CIs were interpreted as evidence
 397 of a significant difference in heritability estimates between models, which provides
 398 a conservative test, as this criterion is more stringent than a direct significance test
 399 at $\alpha=0.05$.

400 Additionally, we computed z-scores to quantify the magnitude of change in
 401 heritability estimates between models, calculated as:

$$Z = \frac{h_1^2 - h_2^2}{\sqrt{SE_1^2 + SE_2^2}},$$

402 where 1 and 2 represent heritability estimates and respective SEs from the two
 403 models being contrasted.

404 3.5 Estimating Dispersion Components with Double Generalized 405 Models

406 We employed double generalized linear models (dglms) to investigate the effect
 407 of SFOH measures on both the mean and the variance of traits. Dglms extend
 408 traditional generalized linear models by simultaneously modeling both the mean
 409 and the dispersion components of the response variable, allowing us to test whether
 410 the variance of the trait differs across levels of predictor variables. We implemented
 411 the `dglm` package in R with a Gaussian distribution for the mean model and the
 412 default log link for the variance model. We fit the following mean and variance
 413 models for 18 traits:

$$\text{Mean: } E(Y) = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Sex} + \beta_3 \text{Race} + \beta_4 \text{Ethnicity} + \sum_{k=1}^p \delta_k \text{SFOH}_k$$

$$\text{Dispersion: } \log\{\text{Var}(Y)\} = \gamma_0 + \gamma_1 \text{Age} + \gamma_2 \text{Sex} + \gamma_3 \text{Race} + \gamma_4 \text{Ethnicity} + \sum_{k=1}^p \eta_k \text{SFOH}_k$$

where $SFOH_k$ represents the k th component of one of two SFOH measures analyzed in separate models: the first 20 MC axes ($p = 20$) or the SSI as a single measure ($p = 1$). To account for multiple comparisons, we applied a hierarchical Bonferroni correction with four tiers of stringency. Each test was assigned a significance level based on the most stringent threshold it passed:

1. **Overall:** $p < 0.05$ / (total number of tests across all traits and SFOH measures)
2. **Family-wise:** $p < 0.05$ / (number of tests within the SFOH measure)
3. **Trait-wise:** $p < 0.05$ / (number of predictors per trait)
4. **Nominal:** $p < 0.05$ (uncorrected)

We then labeled predictors according to the highest significance level achieved (overall, family-wise, trait-wise, or nominal).

3.6 Social Determinants of Health Survey and SFOH Derivation

We derived Social factors of health (SFOH) measures from the All of Us Social Determinants of Health (SDOH) survey (v7), which captures data on factors such as income, education, and housing stability. Our analysis used the survey data as it was available in the v7 dataset. Out of the 397,732 participants eligible for the survey by June 30, 2022, 117,783 (29.6%) had completed it. From these respondents, 94,030 had genotyping array data available. We one-hot encoded the raw survey responses to create a structured dataset for downstream analysis. From this encoded data, we generated two distinct SFOH measures: multiple correspondence analysis axes (MCA), and a Social Similarity Index (SSI), as detailed below.

3.6.1 Survey Preparation

Survey items were organized into themes (i.e. conceptual domains grouping related questions, such as “perceived stress,” “social support,” “loneliness,” etc.) based on the SFOH Task Force’s consensus-based conceptual frameworks. All available survey completers ($n=117,783$) were used to generate the SFOH measures to ensure stable estimates.

3.6.2 MCA

To capture the multidimensional nature of SFOH, we applied dimensionality reduction to one-hot encoded survey data. Our primary analysis used MCA, implemented in Python using the `prince` package, as it is the methodologically appropriate technique for categorical data. We merged the resulting component scores into a single analysis dataset using unique subject identifiers. We include up to the first 40 SFOH MC axes, which capture the major axes of social-factor heterogeneity in our cohort, as linear covariates in downstream analyses.

As a sensitivity analysis to assess the robustness of our conclusions to the choice of dimension reduction technique, we also performed PCA on the same one-hot encoded survey data using `prcomp` in R. Results from PCA-based analyses were qualitatively consistent with the primary MCA findings throughout. Key PCA analyses are presented in the Supplement, and referenced in the main text where relevant.

455 3.6.3 Environmental Relatedness Matrix (ERM)

456 We constructed an environmental relatedness matrix (ERM) from the same one-hot
 457 encoded SFOH survey data. The ERM is conceptually analogous to a GRM, here
 458 we derive it entirely from SFOH survey data. In this approach, every individual's
 459 binary response vector forms a row of the data matrix

$$X \in \{0, 1\}^{n \times p},$$

460 where n is the number of individuals and p is the total number of response-level
 461 columns. To ensure that each question contributes equally regardless of how many
 462 response levels it has, we form a diagonal weight matrix

$$W = \text{diag}(w_1, \dots, w_p),$$

463 with

$$w_\ell = \frac{1}{|\{\text{levels of question corresponding to column } \ell\}|}.$$

464 We define the weighted design matrix

$$X_w = X W,$$

465 so that

$$(X_w)_{i\ell} = w_\ell X_{i\ell}.$$

466 To account for missing responses, we define a binary mask matrix

$$M \in \{0, 1\}^{n \times p},$$

467 by setting

$$M_{i\ell} = \begin{cases} 1, & \text{if individual } i \text{ answered level column } \ell, \\ 0, & \text{otherwise,} \end{cases}$$

468 and its weighted version

$$M_w = M W.$$

469 All matrix operations are carried out on the sparse weighted matrices X_w and M_w .
 470 Although $\text{ERM}_{ii} = 1$ holds by definition, we never explicitly compute or store
 471 diagonal entries.

472 3.6.4 Social Similarity Index (SSI)

473 The SSI provides an individual-level summary of how similar each person's survey
 474 responses are to all other participants in the cohort, accounting for the structure
 475 of the survey instrument. Conceptually, individuals with more similar social cir-
 476 cumstances (as captured by their survey responses) will have higher SSI values,
 477 reflecting shared social environments. The SSI is computed by comparing each per-
 478 son's response pattern to every other person's responses across all SFOH survey
 479 questions, with appropriate weighting to ensure that questions with more response
 480 options do not disproportionately influence the similarity calculation. Mathemat-
 481 ically, this is accomplished through the ERM. Rather than computing or storing

the full $n \times n$ ERM, we summarize each individual’s social similarity to the cohort using the ERM row sums. We define each row sum as

$$s_i = \sum_{j=1}^n \text{ERM}_{ij}.$$

We compute the row sum in $O(np)$ time by first defining the two vectors

$$c = X_w^\top \mathbf{1}_n, \quad d = M^\top \mathbf{1}_n,$$

Here, d gives the unweighted sum of answered items per column. We then compute the individual-level weighted sums

$$u = X_w c, \quad v = M_w d.$$

Each individual’s SSI is then

$$s_i = \frac{u_i}{v_i}, \quad i = 1, \dots, n,$$

and we set any indeterminate $0/0$ to zero (so that individuals with no overlapping responses are treated as unrelated). We standardize the SSI to have mean zero and unit variance. The standardized SSI captures each individual’s overall social similarity to the cohort which we include in downstream regression models to account for shared social context.

4 Discussion

Our results demonstrate that SFOH measures reduce HE regression heritability estimates for a subset of anthropometric and metabolic traits—particularly BMI, hip and waist circumference, and HDL cholesterol—in this multi-ancestry analysis group.

HE regression’s sensitivity to SFOH adjustment demonstrates how social factors may proxy genetic structure, leading to inflated estimates when unmodeled. This makes HE regression a valuable tool for detecting potential confounding. SFOH-associated variance detected by HE regression may represent confounding or indirect genetic effects rather than direct additive genetic variance.

The heritability of SFOH measures themselves provides direct evidence that they capture population structure. Several SFOH measures appeared moderately heritable when adjusting for basic demographics or a limited number of genetic PCs (Fig. S7 and Fig. S8). However, with comprehensive adjustment using seven genetic PCs, these heritability estimates decreased dramatically (Table S8). Measures that remained significantly different than zero converged to a common low baseline (Table S7). This pattern was recapitulated in the European-ancestry subgroup with the recalculated SFOH MCs (Fig. S9), indicating that SFOH measures capture stratification that persists within more ancestrally homogeneous populations. If different SFOH measures reflected distinct genetic influences on different social factors, we would expect divergent estimates after PC adjustment; instead, their convergence indicates they all capture the same underlying confounder. This finding explains why including SFOH measures as covariates reduces trait heritability estimates—they remove stratified variance that would otherwise be misattributed to genetics by HE regression.

518 This finding has important implications for genetic studies. First, it demon-
519 strates that inadequate control for population structure can create spurious evidence
520 of genetic effects on environmental and social factors. Second, it explains why includ-
521 ing SFOH measures as covariates reduces trait heritability estimates—they capture
522 confounding social stratification that would otherwise be misattributed to genetics.
523 Third, it suggests that the small residual heritability ($\approx 1 - 2\%$) observed even with
524 seven PCs likely represents fine-scale population structure rather than meaningful
525 genetic effects on social environments.

526 Furthermore, several SFOH measures that showed this pattern exhibit
527 widespread, though small, effects on trait dispersion even after adjusting for age,
528 sex, race, and ethnicity (Fig. 5), indicating they influence both trait means and
529 variances. Notably, while the SSI showed a similar initial heritability pattern, it
530 does not reduce trait heritability estimates, suggesting that reductions are driven
531 by specific social dimensions rather than global social similarity.

532 The interpretation of the initial apparent heritability of SFOH measures is
533 complex, as it may reflect genuine genetic influence on social factors through gene-
534 environment correlation [31], residual population stratification not fully captured
535 by the first three genetic PCs, technical artifacts from survey data correlation struc-
536 ture, social stratification [32], or environmental confounding [33]—processes that
537 can induce genetic associations with social factors even without direct biological
538 pathways. Our finding that these heritability estimates decrease with comprehen-
539 sive genetic PC adjustment strongly suggests that, in our multi-ancestry cohort,
540 residual population stratification is the primary driver. Regardless of mechanism,
541 these SFOH measures introduce structured variance that can inflate trait heri-
542 tability estimates in HE regression when unmodeled. While adjusting for heritable
543 covariates risks removing true genetic signal [34], our results indicate that in this
544 context, SFOH adjustment primarily removes stratification that would otherwise
545 be misattributed to genetics.

546 We acknowledge several methodological limitations. First, the comprehensive
547 dispersion models include up to 33 predictors and may capture sample-specific
548 patterns that do not generalize; these results should be viewed as hypothesis-
549 generating rather than definitive. Second, while our multi-ancestry design limits
550 direct generalization to any single population group, it provides advantages for
551 detecting population stratification. The pronounced genetic structure in multi-
552 ancestry cohorts makes SFOH-ancestry correlations more detectable, suggesting our
553 findings may represent a conservative bound on confounding that could be sub-
554 tler but still present in more homogeneous populations. Third, while our use of
555 seven genetic PCs was sufficient to demonstrate that SFOH heritability estimates
556 decrease, even this number may not capture all fine-scale population structure.
557 Residual stratification could therefore still be partially captured by SFOH mea-
558 sures or influence trait heritability estimates. Fourth, our use of self-identified
559 race—a social construct that imperfectly captures both genetic ancestry and social
560 experiences—risks conflating genetic background with socially mediated SFOH
561 exposures; between group comparisons are not interpretable as true population
562 differences and are indistinguishable from statistical artifacts in this study.

563 These results highlight that SFOH structure can meaningfully alter SNP-
564 based heritability estimates and trait variance patterns. A small number of social
565 dimensions—social support, neighborhood environment, feelings of discrimination,

566 stress, and spirituality—appear especially important in shaping the variance of com-
567 plex traits. The convergence of SFOH heritability estimates across diverse social
568 dimensions provides evidence that these measures primarily capture population
569 stratification rather than independent genetic effects on social environments. This
570 work underscores that for traits with strong social patterning, particularly anthro-
571 pometric measures, measuring and adjusting for specific social environmental factors
572 may be necessary to avoid misattributing social influences to genetic effects. Future
573 work should focus on methods to disentangle the roles of genetic structure, social
574 environments, and their interactions in shaping trait variance.

575 5 Data availability

576 Individual-level genotype and survey data used in this study are available to regis-
577 tered researchers through the All of Us Researcher Workbench ([https://workbench.
578 researchallofus.org/](https://workbench.researchallofus.org/)) following Data and Research Center (DRC) access procedures.
579 Genomic data (v7 array release) and survey responses, including the Social Deter-
580 minants of Health survey, are part of the All of Us [35] controlled tier dataset. Other
581 information is available upon request.

582 **Acknowledgments.** We thank the All of Us Research Program and its partici-
583 pants. We thank Lily Hamilton for thoughtful feedback that led to the improve-
584 ment of this manuscript. This research was supported by NHGRI CEGS RM1:
585 RM1HG010461 (NSA) and NIH NCI U54CA267738 (SAM), The Cornell University
586 Graduate School Deans Excellence Fellowship (OSRA), and The Cornell University
587 Graduate School Walter Schonlenk and Leister Fellowships (JLL).

588 **Declarations.** The authors declare no competing interests.

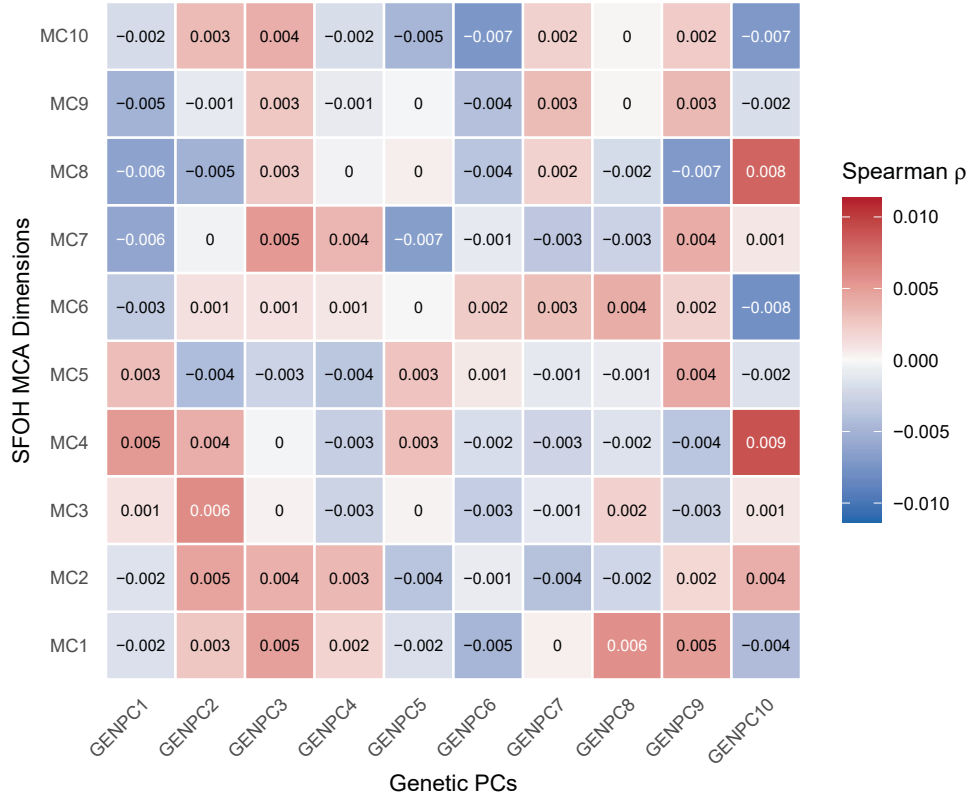


Fig. 1: Relationship between genetic structure and SFOH measures. Spearman correlations between the 10 genetic PCs (x-axis) and top 10 SFOH survey derived MC axes (y-axis).

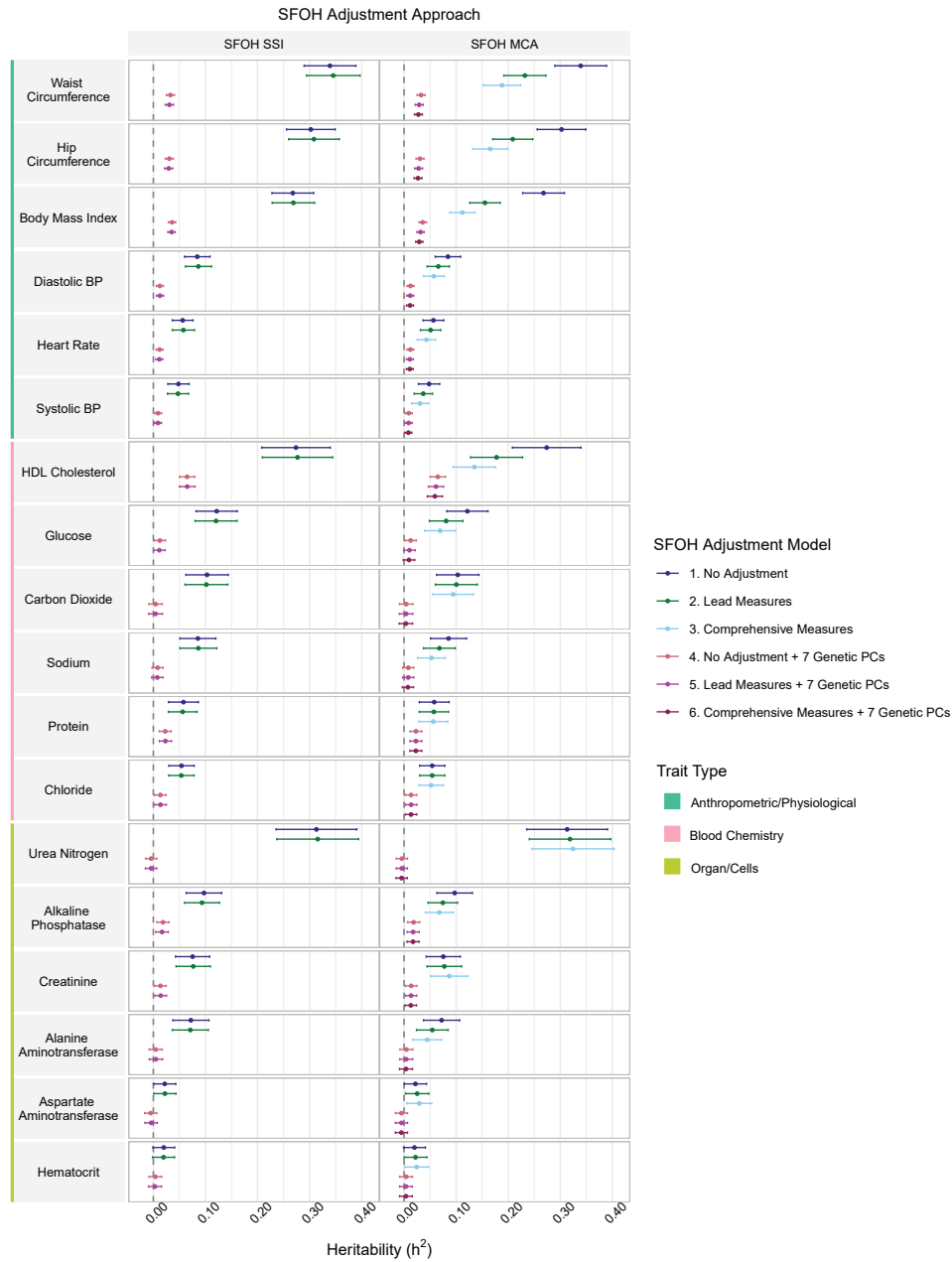


Fig. 2: Combined analysis group Haseman-Elston (HE) regression across SFOH-covariate models. HE regression heritability estimates for all relevant participants with 95% confidence intervals for 18 traits (rows), calculated using two Social Factors of Health (SFOH) summarization strategies (columns). All models are adjusted for age, sex, race, and ethnicity. Models with SFOH adjustment include lead (MCs 1:6 or SSI) or comprehensive (MCs 1:40) adjustment. Adding SFOH MC axes (models 2,3) reduces heritability estimates for body mass index, HDL cholesterol, hip circumference, glucose, and waist circumference relative to the base model (model 1). Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero.

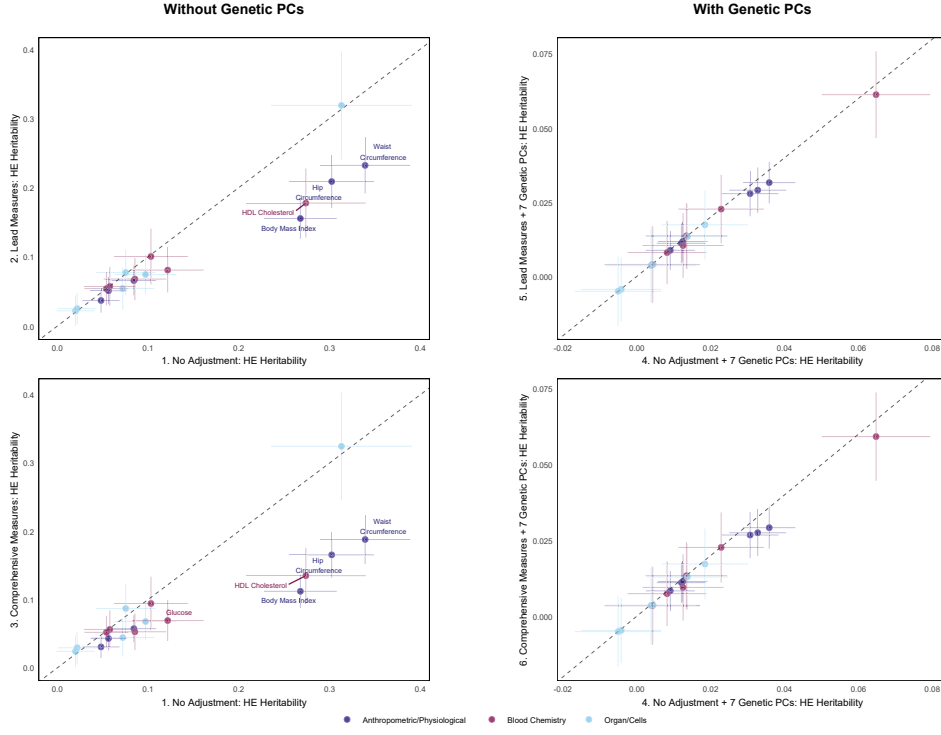


Fig. 3: Combined analysis group Haseman-Elston (HE) regression across SFOH MCA adjustment models. HE regression heritability estimates for all relevant participants with 95% confidence intervals for 18 traits, calculated using SFOH MCA. Panels are arranged in a 2x2 grid: models without genetic PCs (left) and with genetic PCs (right). Each panel shows a pairwise model comparison with model numbers noted at the beginning of the x- and y- axis labels. All models are adjusted for age, sex, race, and ethnicity. Models with SFOH adjustment include lead (MCs 1:6) or comprehensive (MCs 1:40) adjustment. Adding SFOH MC axes (models 2,3) reduces heritability estimates for BMI, HDL cholesterol, hip circumference, glucose, and waist circumference relative to the base model (model 1). Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates the $y = x$ line.

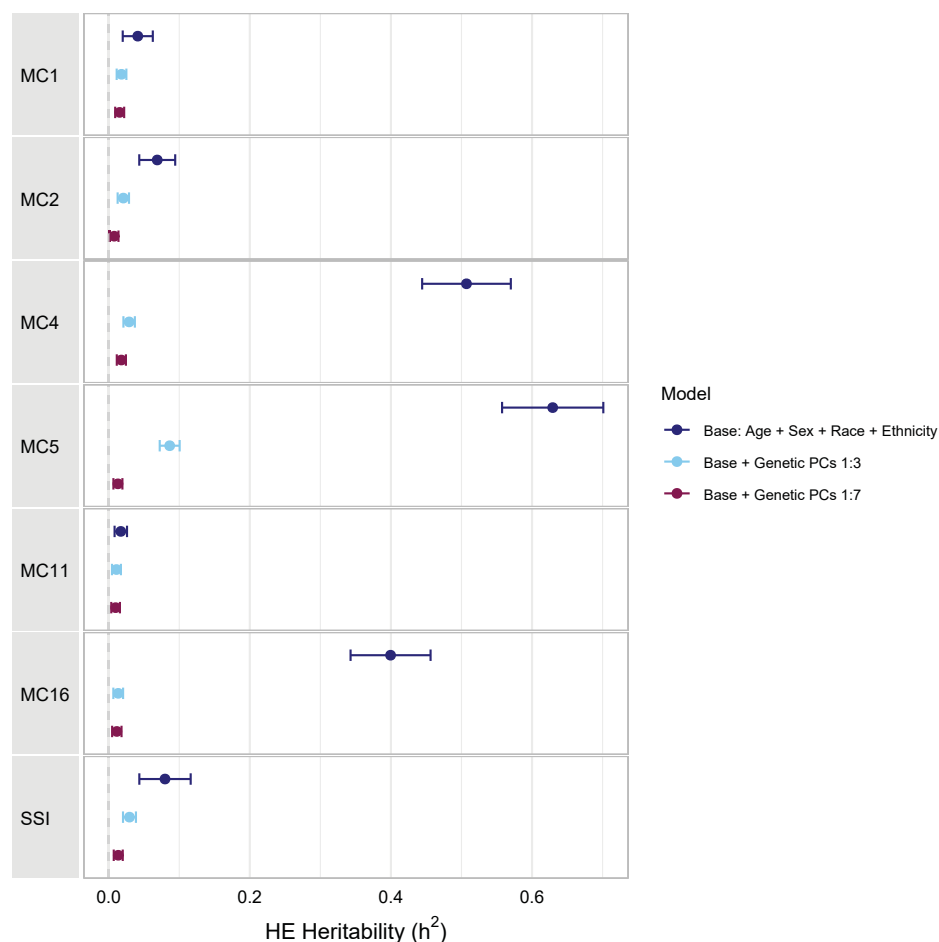


Fig. 4: Combined analysis group Haseman-Elston (HE) regression heritability of SFOH measures that remain heritable after including seven genetic PCs. HE regression heritability estimates for all relevant participants with 95% confidence intervals for SFOH measures that appear heritable after comprehensive genetic PC adjustment. Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero. HE regression heritability estimates show progressive reduction with genetic PC adjustment, demonstrating that apparent heritability primarily reflects population stratification.

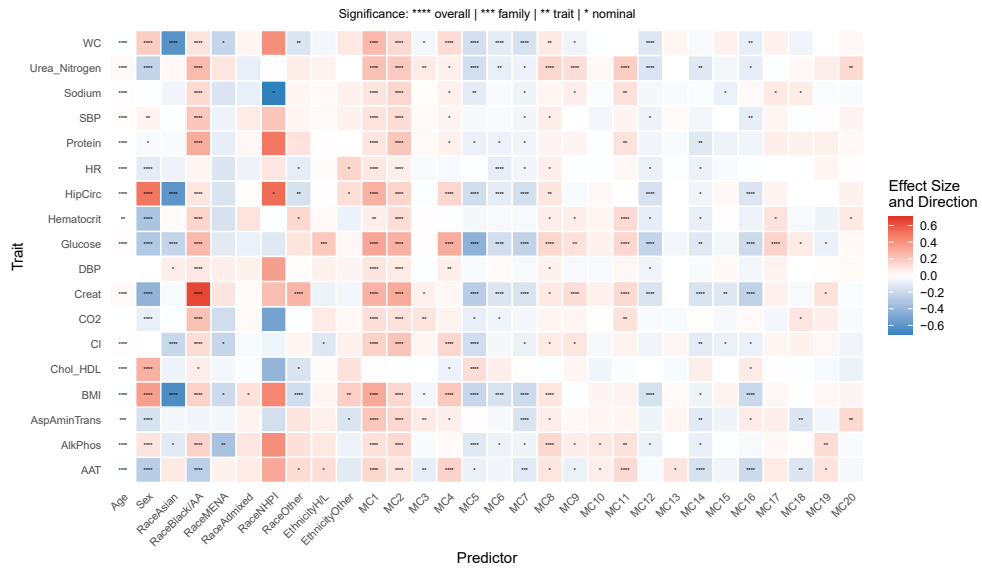


Fig. 5: SFOH MC axes dispersion effects. Heatmap of significant predictors in trait (y-axis) dispersion models with demographics and SFOH MC axes (x-axis). The color gradient (blue to red) and intensity indicates the direction and magnitude of the effect size of the measure on the variance of each trait. Significance is based on a hierarchical Bonferroni correction with four tiers of stringency. Each test was assigned a significance level based on the most stringent threshold it passed and is indicated with “*”.

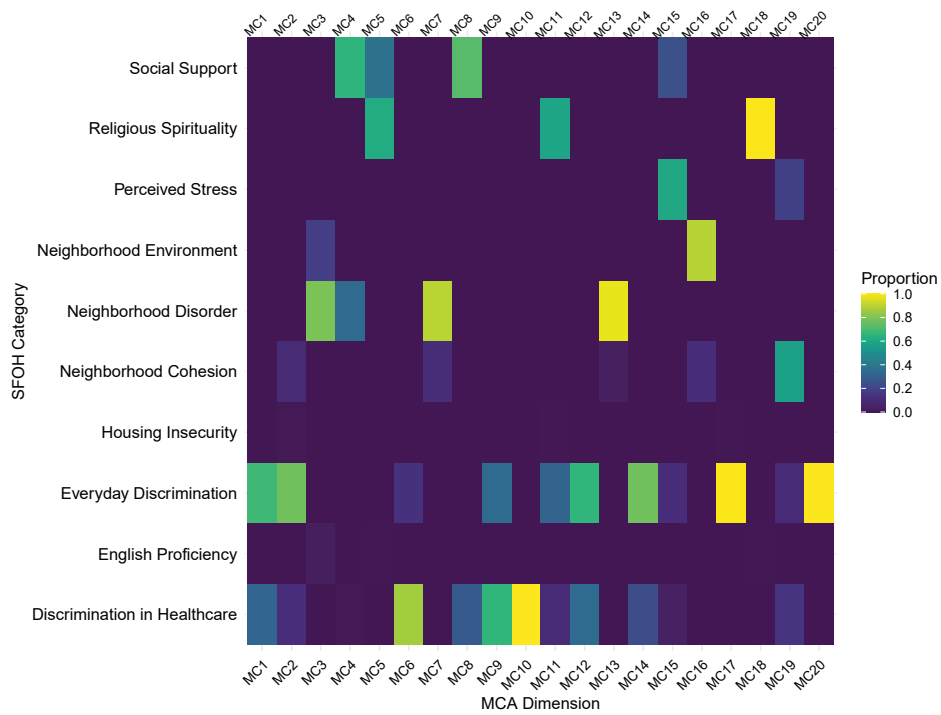


Fig. 6: SFOH MCA dimension topic loadings heatmap. This heatmap depicts the contribution of each survey topic (y-axis) to SFOH MC axes (x-axis) derived from the All of Us Social Determinants of Health Survey. The color gradient (purple to yellow) represents each topic's proportional weight, indicating its relative importance in defining each component. Each column sums to 1.

589 References

- 590 [1] Emil Uffelmann, Qin Qin Huang, Nchangwi Syntia Munung, Jantina de Vries,
591 Yukinori Okada, Alicia R. Martin, Hilary C. Martin, Tuuli Lappalainen,
592 and Danielle Posthuma. Genome-wide association studies. *Nature Reviews*
593 *Methods Primers*, 1(1):59, August 2021. ISSN 2662-8449. doi: 10.1038/
594 s43586-021-00056-9. Publisher: Nature Publishing Group.
- 595 [2] Jonathan Marchini, Lon R. Cardon, Michael S. Phillips, and Peter Donnelly.
596 The effects of human population structure on large genetic association studies.
597 *Nature Genetics*, 36(5):512–517, May 2004. ISSN 1546-1718. doi: 10.1038/
598 ng1337. Publisher: Nature Publishing Group.
- 599 [3] John Novembre, Toby Johnson, Katarzyna Bryc, Zoltán Kutalik, Adam R.
600 Boyko, Adam Auton, Amit Indap, Karen S. King, Sven Bergmann, Matthew R.
601 Nelson, Matthew Stephens, and Carlos D. Bustamante. Genes mirror geog-
602 raphy within Europe. *Nature*, 456(7218):98–101, November 2008. ISSN
603 1476-4687. doi: 10.1038/nature07331. Publisher: Nature Publishing Group.
- 604 [4] Daniel John Lawson, Neil Martin Davies, Simon Haworth, Bilal Ashraf, Lau-
605 rence Howe, Andrew Crawford, Gibran Hemani, George Davey Smith, and
606 Nicholas John Timpson. Is population structure in the genetic biobank era
607 irrelevant, a challenge, or an opportunity? *Human Genetics*, 139(1):23–41,
608 January 2020. ISSN 1432-1203. doi: 10.1007/s00439-019-02014-8.
- 609 [5] Kevin H. Nguyen and Megan B. Cole. Social Risk Factors, Health Insurance
610 Coverage, and Inequities in Access to Care. *American Journal of Preventive*
611 *Medicine*, 68(1):145–153, January 2025. ISSN 0749-3797, 1873-2607. doi: 10.
612 1016/j.amepre.2024.09.005. Publisher: Elsevier.
- 613 [6] Mohammad Hashim Jilani, Zulqarnain Javed, Tamer Yahya, Javier Valero-
614 Elizondo, Safi U. Khan, Bita Kash, Ron Blankstein, Salim S. Virani, Michael J.
615 Blaha, Prachi Dubey, Adnan A. Hyder, Farhaan S. Vahidy, Miguel Cainzos-
616 Achirica, and Khurram Nasir. Social Determinants of Health and Cardiovascu-
617 lar Disease: Current State and Future Directions Towards Healthcare Equity.
618 *Current Atherosclerosis Reports*, 23(9):55, July 2021. ISSN 1534-6242. doi:
619 10.1007/s11883-021-00949-w.
- 620 [7] Cynthia A. Gómez, Dushanka V. Kleinman, Nico Pronk, Glenda L. Wrenn Gor-
621 don, Emmeline Ochiai, Carter Blakey, Ayanna Johnson, and Karen H.
622 Brewer. Addressing Health Equity and Social Determinants of Health Through
623 Healthy People 2030. *Journal of Public Health Management and Practice*, 27
624 (Supplement 6):S249, December 2021. ISSN 1078-4659. doi: 10.1097/PHH.
625 0000000000001297.
- 626 [8] Yarden S. Fraiman and Monica H. Wojcik. The influence of social determinants
627 of health on the genetic diagnostic odyssey: who remains undiagnosed, why,
628 and to what effect? *Pediatric Research*, 89(2):295–300, January 2021. ISSN
629 1530-0447. doi: 10.1038/s41390-020-01151-5. Publisher: Nature Publishing
630 Group.

- [9] Jimmy Phuong, Naomi O. Riches, Charisse Madlock-Brown, Deborah Duran, Luca Calzoni, Juan C. Espinoza, Gora Datta, Ramakanth Kavuluru, Nicole G. Weiskopf, Cavin K. Ward-Caviness, and Asiyah Yu Lin. Social Determinants of Health Factors for Gene–Environment COVID-19 Research: Challenges and Opportunities. *Advanced Genetics*, 3(2): 2100056, 2022. ISSN 2641-6573. doi: 10.1002/ggn2.202100056. eprint: <https://advanced.onlinelibrary.wiley.com/doi/pdf/10.1002/ggn2.202100056>.
- [10] The “All of Us” Research Program | New England Journal of Medicine.
- [11] Hailan Liu and Guo-Bo Chen. A novel genomic prediction method combining randomized Haseman-Elston regression with a modified algorithm for Proven and Young for large genomic data. *The Crop Journal*, 10(2):550–554, April 2022. ISSN 2214-5141. doi: 10.1016/j.cj.2021.09.001.
- [12] Ali Pazokitoroudi, Zhengtong Liu, Andrew Dahl, Noah Zaitlen, Saharon Rosset, and Sriram Sankararaman. A scalable and robust variance components method reveals insights into the architecture of gene-environment interactions underlying complex traits. *The American Journal of Human Genetics*, 111(7): 1462–1480, July 2024. ISSN 0002-9297, 1537-6605. doi: 10.1016/j.ajhg.2024.05.015. Publisher: Elsevier.
- [13] Suresh K. Bhavnani, Weibin Zhang, Daniel Bao, Mukaila Raji, Veronica Ajewole, Rodney Hunter, Yong-Fang Kuo, Susanne Schmidt, Monique R. Pappadis, Elise Smith, Alex Bokov, Timothy Reistetter, Shyam Visweswaran, and Brian Downer. Subtyping Social Determinants of Health in All of Us: Network Analysis and Visualization Approach. *medRxiv*, page 2023.01.27.23285125, August 2023. doi: 10.1101/2023.01.27.23285125.
- [14] Sonali Gupta, Vincent Lam, I. King Jordan, and Leonardo Mariño-Ramírez. A composite socioeconomic deprivation index from All of Us survey data: associations with health outcomes and disparities. *medRxiv*, page 2024.10.04.24314904, October 2024. doi: 10.1101/2024.10.04.24314904.
- [15] Jane Brown, Abhijith Biji, Kathleen Ferar, Vikas Pejaver, Eimear E. Kenny, Bian Liu, and Samira Asgari. A Scalable Framework to Integrate Social Determinants of Health into Disease Risk Models using Biobank Survey Data, September 2025. ISSN: 3067-2007 Pages: 2025.09.29.25336903.
- [16] Cole Brokamp, Andrew F. Beck, Neera K. Goyal, Patrick Ryan, James M. Greenberg, and Eric S. Hall. Material community deprivation and hospital utilization during the first year of life: an urban population–based cohort study. *Annals of Epidemiology*, 30:37–43, February 2019. ISSN 1047-2797. doi: 10.1016/j.annepidem.2018.11.008.
- [17] Arun Durvasula and Alkes L. Price. Distinct explanations underlie gene-environment interactions in the UK Biobank. *The American Journal of Human Genetics*, 112(3):644–658, March 2025. ISSN 0002-9297. doi: 10.1016/j.ajhg.2025.01.014. Publisher: Elsevier.

- [18] Gordon K Smyth. Adjusted Likelihood Methods for Modelling Dispersion in Generalized Linear Models. *Environmetricsr*, 1999.
- [19] Stefania Benonisdottir and Augustine Kong. Studying the genetics of participation using footprints left on the ascertained genotypes. *Nature Genetics*, 55(8): 1413–1420, August 2023. ISSN 1546-1718. doi: 10.1038/s41588-023-01439-2. Publisher: Nature Publishing Group.
- [20] Gianmarco Mignogna, Caitlin E. Carey, Robbee Wedow, Nikolas Baya, Matia Cordioli, Nicola Pirastu, Rino Bellocco, Kathryn Fiuza Malerbi, Michel G. Nivard, Benjamin M. Neale, Raymond K. Walters, and Andrea Ganna. Patterns of item nonresponse behaviour to survey questionnaires are systematic and associated with genetic loci. *Nature Human Behaviour*, 7(8):1371–1387, August 2023. ISSN 2397-3374. doi: 10.1038/s41562-023-01632-7.
- [21] Shaila Musharoff, Danny Park, Andy Dahl, Joshua Galanter, Xuanyao Liu, Scott Huntsman, Celeste Eng, Esteban G. Burchard, Julien F. Ayroles, and Noah Zaitlen. Existence and implications of population variance structure, October 2018. Pages: 439661 Section: New Results.
- [22] Tesfaye B. Mersha and Tilahun Abebe. Self-reported race/ethnicity in the age of genomic research: its potential impact on understanding health disparities. *Human Genomics*, 9(1):1, January 2015. ISSN 1479-7364. doi: 10.1186/s40246-014-0023-x.
- [23] Roseann E. Peterson, Karoline Kuchenbaecker, Raymond K. Walters, Chia-Yen Chen, Alice B. Popejoy, Sathish Periyasamy, Max Lam, Conrad Iyegbe, Rona J. Strawbridge, Leslie Brick, Caitlin E. Carey, Alicia R. Martin, Jacquelyn L. Meyers, Jinni Su, Junfang Chen, Alexis C. Edwards, Allan Kalungi, Nastassja Koen, Lerato Majara, Emanuel Schwarz, Jordan W. Smoller, Eli A. Stahl, Patrick F. Sullivan, Evangelos Vassos, Bryan Mowry, Miguel L. Prieto, Alfredo Cuellar-Barboza, Tim B. Bigdeli, Howard J. Edenberg, Hailiang Huang, and Laramie E. Duncan. Genome-wide Association Studies in Ancestrally Diverse Populations: Opportunities, Methods, Pitfalls, and Recommendations. *Cell*, 179(3):589–603, October 2019. ISSN 0092-8674, 1097-4172. doi: 10.1016/j.cell.2019.08.051. Publisher: Elsevier.
- [24] Joshua G. Schraiber and Michael D. Edge. Heritability within groups is uninformative about differences among groups: Cases from behavioral, evolutionary, and statistical genetics. *Proceedings of the National Academy of Sciences*, 121(12):e2319496121, March 2024. doi: 10.1073/pnas.2319496121. Publisher: Proceedings of the National Academy of Sciences.
- [25] Zhaotong Lin, Souvik Seal, and Saonli Basu. Estimating SNP heritability in presence of population substructure in biobank-scale datasets. *Genetics*, 220(4):iyac015, April 2022. ISSN 1943-2631. doi: 10.1093/genetics/iyac015.
- [26] Jian Yang, S. Hong Lee, Michael E. Goddard, and Peter M. Visscher. GCTA: A Tool for Genome-wide Complex Trait Analysis. *American Journal of Human Genetics*, 88(1):76–82, January 2011. ISSN 0002-9297. doi: 10.1016/j.ajhg.

2010.11.011.

- [27] J. K. Haseman and R. C. Elston. The investigation of linkage between a quantitative trait and a marker locus. *Behavior Genetics*, 2(1):3–19, March 1972. ISSN 1573-3297. doi: 10.1007/BF01066731.
- [28] Mateus H. Gouveia, Karlijn A. C. Meeks, Victor Borda, Thiago P. Leal, Fernanda S. G. Kehdy, Reagan Mogire, Ayo P. Doumatey, Eduardo Tarazona-Santos, Rick A. Kittles, Ignacio F. Mata, Timothy D. O’Connor, Adebawale A. Adeyemo, Daniel Shriner, and Charles N. Rotimi. Subcontinental genetic variation in the All of Us Research Program: Implications for biomedical research. *American Journal of Human Genetics*, 112(6):1286–1301, June 2025. ISSN 1537-6605. doi: 10.1016/j.ajhg.2025.04.012.
- [29] Second-generation PLINK: rising to the challenge of larger and richer datasets | GigaScience | Oxford Academic.
- [30] Ani Manichaikul, Josyf C. Mychaleckyj, Stephen S. Rich, Kathy Daly, Michèle Sale, and Wei-Min Chen. Robust relationship inference in genome-wide association studies. *Bioinformatics*, 26(22):2867–2873, November 2010. ISSN 1367-4803. doi: 10.1093/bioinformatics/btq559.
- [31] Abdel Abdellaoui, Conor V. Dolan, Karin J. H. Verweij, and Michel G. Nivard. Gene–environment correlations across geographic regions affect genome-wide association studies. *Nature Genetics*, 54(9):1345–1354, Sep 2022. ISSN 1546-1718. doi: 10.1038/s41588-022-01158-0.
- [32] Abdel Abdellaoui, David Hugh-Jones, Loic Yengo, Kathryn E. Kemper, Michel G. Nivard, Laura Veul, Yan Holtz, Brendan P. Zietsch, Timothy M. Frayling, Naomi R. Wray, Jian Yang, Karin J. H. Verweij, and Peter M. Visscher. Genetic correlates of social stratification in great britain. *Nature Human Behaviour*, 3(12):1332–1342, Dec 2019. ISSN 2397-3374. doi: 10.1038/s41562-019-0757-5.
- [33] Laurence J. Howe, Michel G. Nivard, Tim T. Morris, Ailin F. Hansen, Humaira Rasheed, Yoonsu Cho, Geetha Chittoor, Rafael Ahlskog, Penelope A. Lind, Teemu Palviainen, Matthijs D. van der Zee, Rosa Cheesman, Massimo Mangino, Yunzhang Wang, Shuai Li, Lucija Klaric, Scott M. Ratliff, Lawrence F. Bielak, Marianne Nygaard, Alexandros Giannelis, Emily A. Willoughby, Chandra A. Reynolds, Jared V. Balbona, Ole A. Andreassen, Helga Ask, Aris Baras, Christopher R. Bauer, Dorret I. Boomsma, Archie Campbell, Harry Campbell, Zhengming Chen, Paraskevi Christofidou, Elizabeth Corfield, Christina C. Dahm, Deepika R. Dokuru, Luke M. Evans, Eco J. C. de Geus, Sudheer Giddaluru, Scott D. Gordon, K. Paige Harden, W. David Hill, Amanda Hughes, Shona M. Kerr, Yongkang Kim, Hyeokmoon Kweon, Antti Latvala, Deborah A. Lawlor, Liming Li, Kuang Lin, Per Magnus, Patrik K. E. Magnusson, Travis T. Mallard, Pekka Martikainen, Melinda C. Mills, Pål Rasmus Njølstad, John D. Overton, Nancy L. Pedersen, David J. Porteous, Jeffrey Reid, Karri Silventoinen, Melissa C. Southey, Camilla Stoltenberg, Elliot M. Tucker-Drob, Margaret J. Wright, Philipp D. Koellinger, Daniel J. Benjamin,

- 757 Patrick Turley, John K. Hewitt, Matthew C. Keller, Michael C. Stallings,
758 James J. Lee, Kaare Christensen, Sharon L. R. Kardia, Patricia A. Peyser, Jen-
759 nifer A. Smith, James F. Wilson, John L. Hopper, Sara Hägg, Tim D. Spector,
760 Jean-Baptiste Pingault, Robert Plomin, Alexandra Havdahl, Meike Bartels,
761 Nicholas G. Martin, Sven Oskarsson, Anne E. Justice, Iona Y. Millwood, Kris-
762 tian Hveem, Øyvind Naess, Cristen J. Willer, Bjørn Olav Åsvold, Jaakko
763 Kaprio, Sarah E. Medland, Robin G. Walters, David M. Evans, George Davey
764 Smith, Caroline Hayward, Ben Brumpton, Gibran Hemani, Neil M. Davies,
765 George Davey Smith, Social Science Genetic Association Consortium, and
766 Within Family Consortium. Within-sibship genome-wide association analyses
767 decrease bias in estimates of direct genetic effects. *Nature Genetics*, 54(5):
768 581–592, May 2022. ISSN 1546-1718. doi: 10.1038/s41588-022-01062-7.
- 769 [34] Hugues Aschard, Bjarni J. Vilhjálmsson, Amit D. Joshi, Alkes L. Price, and
770 Peter Kraft. Adjusting for Heritable Covariates Can Bias Effect Estimates in
771 Genome-Wide Association Studies. *American Journal of Human Genetics*, 96
772 (2):329–339, February 2015. ISSN 0002-9297. doi: 10.1016/j.ajhg.2014.12.021.
- 773 [35] The All of Us Research Program Genomics Investigators. Genomic data in the
774 All of Us Research Program. *Nature*, 627(8003):340–346, March 2024. ISSN
775 1476-4687. doi: 10.1038/s41586-023-06957-x. Publisher: Nature Publishing
776 Group.

Table S1: Sample size of females and males that have genetic data and responded to the SFOH survey.

	Sex	Count
1	Female	54967
2	Male	30996

Table S2: Sample size of each self-identified race group that has genetic data and responded to the SFOH survey.

Self-Identified Race	Count
Black or African American	6513
Asian	2318
Middle Eastern or North African	380
More than one population	1511
Native Hawaiian or Other Pacific Islander	42
White	67164
Other	8035

Table S3: Overall Missingness by Theme

Theme	Total Responses	Missing	% Missing
Religious Attendance	88016	10716	12.2
Neighborhood Environment	704128	51475	7.3
Housing Instability	88016	4594	5.2
Perceived Stress	880160	37119	4.2
Neighborhood Disorder	1144208	34943	3.1
Housing Insecurity	88016	2641	3.0
Discrimination in Healthcare Settings	616112	17083	2.8
Everyday Discrimination	792144	20631	2.6
Social Support	704128	15945	2.3
Neighborhood Cohesion	352064	7227	2.1
Loneliness	704128	14207	2.0
Religious Spirituality	528096	8518	1.6
Food Insecurity	176032	1622	0.9

Table S4: Percentage of question level non-response by “Theme” and Self-Identified Race.
Rows containing less than 20 individuals removed as per All of Us data usage agreement.

Theme	Race	Total Responses	Missing (%)
Discrimination in Healthcare	Black or African American	46081	2079 (4.5%)
Discrimination in Healthcare	Other	67312	2909 (4.3%)
Discrimination in Healthcare	Middle Eastern or North African	2681	88 (3.3%)
Discrimination in Healthcare	White	472815	11513 (2.4%)
Discrimination in Healthcare	Asian	16261	299 (1.8%)
Discrimination in Healthcare	More than one population	10654	187 (1.8%)
Everyday Discrimination	Black or African American	59247	2621 (4.4%)
Everyday Discrimination	Other	86544	3538 (4.1%)
Everyday Discrimination	White	607905	13792 (2.3%)
Everyday Discrimination	Asian	20907	383 (1.8%)
Everyday Discrimination	More than one population	13698	223 (1.6%)
Everyday Discrimination	Middle Eastern or North African	3447	55 (1.6%)
Food Insecurity	Black or African American	13166	274 (2.1%)
Food Insecurity	Other	19232	339 (1.8%)
Food Insecurity	Asian	4646	52 (1.1%)
Food Insecurity	White	135090	927 (0.7%)
Housing Insecurity	Black or African American	6583	387 (5.9%)
Housing Insecurity	Other	9616	518 (5.4%)
Housing Insecurity	Asian	2323	60 (2.6%)
Housing Insecurity	White	67545	1625 (2.4%)
Housing Insecurity	More than one population	1522	29 (1.9%)
Housing Instability	White	67545	3668 (5.4%)
Housing Instability	Other	9616	506 (5.3%)
Housing Instability	Black or African American	6583	259 (3.9%)
Housing Instability	Asian	2323	90 (3.9%)
Housing Instability	More than one population	1522	48 (3.2%)
Loneliness	Black or African American	52664	1865 (3.5%)
Loneliness	Other	76928	2698 (3.5%)
Loneliness	Middle Eastern or North African	3064	53 (1.7%)
Loneliness	White	540360	9128 (1.7%)
Loneliness	Asian	18584	292 (1.6%)
Loneliness	More than one population	12176	162 (1.3%)
Neighborhood Cohesion	Black or African American	26332	833 (3.2%)
Neighborhood Cohesion	Other	38464	1155 (3.0%)
Neighborhood Cohesion	White	270180	5063 (1.9%)
Neighborhood Cohesion	Middle Eastern or North African	1532	22 (1.4%)
Neighborhood Cohesion	More than one population	6088	65 (1.1%)
Neighborhood Cohesion	Asian	9292	81 (0.9%)
Neighborhood Disorder	Other	125008	5532 (4.4%)
Neighborhood Disorder	Black or African American	85579	3776 (4.4%)
Neighborhood Disorder	White	878085	24622 (2.8%)
Neighborhood Disorder	Middle Eastern or North African	4979	131 (2.6%)
Neighborhood Disorder	Asian	30199	542 (1.8%)
Neighborhood Disorder	More than one population	19786	319 (1.6%)
Neighborhood Environment	Black or African American	52664	4794 (9.1%)
Neighborhood Environment	Other	76928	6401 (8.3%)
Neighborhood Environment	Middle Eastern or North African	3064	220 (7.2%)
Neighborhood Environment	White	540360	38320 (7.1%)
Neighborhood Environment	More than one population	12176	726 (6.0%)
Neighborhood Environment	Asian	18584	982 (5.3%)

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Table S4 – continued from previous page

Theme	Race	Total Responses	Missing (%)
Perceived Stress	Black or African American	65830	3945 (6.0%)
Perceived Stress	Other	96160	5316 (5.5%)
Perceived Stress	Middle Eastern or North African	3830	173 (4.5%)
Perceived Stress	White	675450	26568 (3.9%)
Perceived Stress	Asian	23230	728 (3.1%)
Perceived Stress	More than one population	15220	369 (2.4%)
Religious Attendance	Black or African American	6583	942 (14.3%)
Religious Attendance	White	67545	8405 (12.4%)
Religious Attendance	Middle Eastern or North African	383	46 (12.0%)
Religious Attendance	Other	9616	1039 (10.8%)
Religious Attendance	More than one population	1522	114 (7.5%)
Religious Attendance	Asian	2323	165 (7.1%)
Religious Spirituality	Black or African American	39498	1127 (2.9%)
Religious Spirituality	Other	57696	1502 (2.6%)
Religious Spirituality	Middle Eastern or North African	2298	58 (2.5%)
Religious Spirituality	White	405270	5522 (1.4%)
Religious Spirituality	More than one population	9132	120 (1.3%)
Religious Spirituality	Asian	13938	180 (1.3%)
Social Support	Black or African American	52664	1987 (3.8%)
Social Support	Other	76928	2744 (3.6%)
Social Support	Middle Eastern or North African	3064	94 (3.1%)
Social Support	White	540360	10638 (2.0%)
Social Support	Asian	18584	333 (1.8%)
Social Support	More than one population	12176	144 (1.2%)

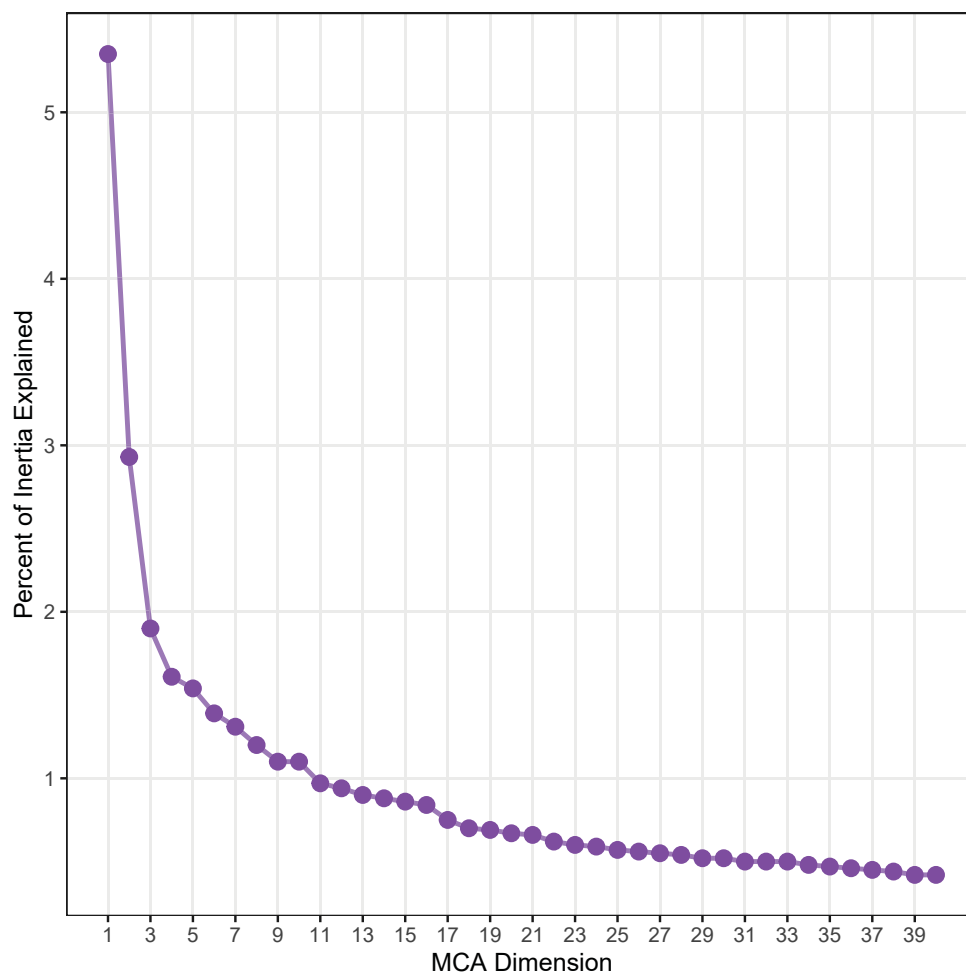


Fig. S1: SFOH MCA scree plot. The lead measures, SFOH MC axes 1:6, account for 14.73% of variance. Comprehensive measures, SFOH MC axes 1:40, account for 38.02% of variance.

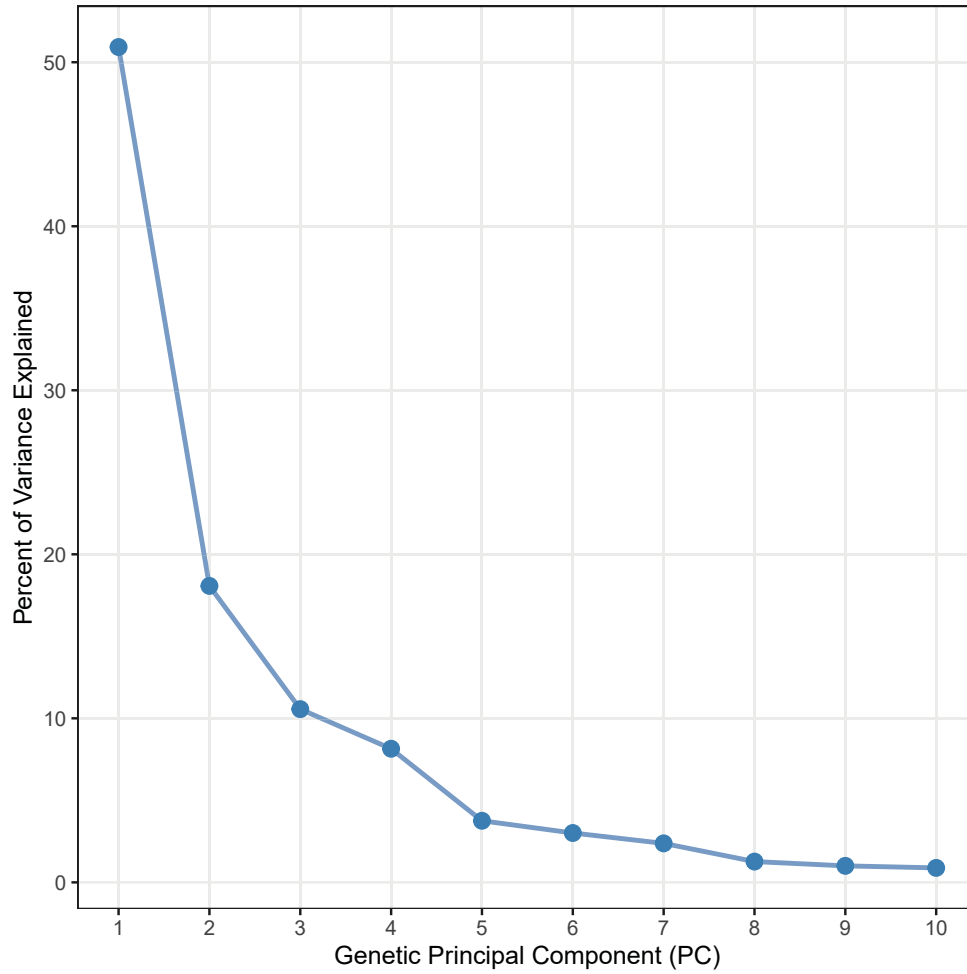


Fig. S2: Genetic PCA scree plot. Genetic PCs 1:3 account for 79.56% of variance, while genetic PCs 1:7 account for 96.84% of variance.



Fig. S3: Female Haseman-Elston (HE) regression across SFOH-covariate models. HE regression heritability estimates of female participants with 95% confidence intervals for 18 traits (rows), calculated using two Social Factors of Health (SFOH) summarization strategies (columns). All models are adjusted for age, race, and ethnicity. Models with SFOH adjustment include lead (MCs 1:6 or SSI) or comprehensive (MCs 1:40) adjustment. Patterns are consistent with the combined analysis group (Fig. 2). Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero.



Fig. S4: Male Haseman-Elston (HE) regression across SFOH-covariate models. HE regression heritability estimates of male participants with 95% confidence intervals for 18 traits (rows), calculated using two Social Factors of Health (SFOH) summarization strategies (columns). All models are adjusted for age, race, and ethnicity. Models with SFOH adjustment include lead (MCs 1:6 or SSI) or comprehensive (MCs 1:40) adjustment. Patterns are consistent with the combined analysis group (Fig. 2). Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero.

Table S5: Haseman–Elston heritability estimates across SFOH SSI adjustment models.

Trait	Model	h^2	SE	95% CI	p
Anthropometric and Physiological Measures					
Waist Circumference	1. No Adjustment	0.339	0.025	[0.289, 0.389]	8.5e-41
Waist Circumference	2. Lead Measures	0.345	0.026	[0.294, 0.397]	1.2e-39
Waist Circumference	4. No Adjustment + 7 Genetic PCs	0.033	0.004	[0.025, 0.040]	8.4e-17
Waist Circumference	5. Lead Measures + 7 Genetic PCs	0.031	0.004	[0.023, 0.039]	1.2e-14
Hip Circumference	1. No Adjustment	0.302	0.024	[0.256, 0.349]	7.1e-37
Hip Circumference	2. Lead Measures	0.308	0.025	[0.260, 0.357]	1.3e-35
Hip Circumference	4. No Adjustment + 7 Genetic PCs	0.031	0.004	[0.023, 0.038]	4.8e-15
Hip Circumference	5. Lead Measures + 7 Genetic PCs	0.029	0.004	[0.022, 0.037]	1.5e-13
Body Mass Index	1. No Adjustment	0.268	0.020	[0.228, 0.308]	3.7e-39
Body Mass Index	2. Lead Measures	0.269	0.021	[0.228, 0.310]	2.8e-38
Body Mass Index	4. No Adjustment + 7 Genetic PCs	0.036	0.004	[0.029, 0.043]	2.0e-23
Body Mass Index	5. Lead Measures + 7 Genetic PCs	0.035	0.004	[0.028, 0.042]	9.9e-22
Diastolic BP	1. No Adjustment	0.084	0.012	[0.060, 0.109]	1.4e-11
Diastolic BP	2. Lead Measures	0.086	0.013	[0.061, 0.112]	1.5e-11
Diastolic BP	4. No Adjustment + 7 Genetic PCs	0.012	0.003	[0.006, 0.019]	0.00028
Diastolic BP	5. Lead Measures + 7 Genetic PCs	0.013	0.003	[0.006, 0.020]	0.00026
Heart Rate	1. No Adjustment	0.056	0.010	[0.036, 0.076]	2.9e-08
Heart Rate	2. Lead Measures	0.058	0.011	[0.036, 0.079]	1.2e-07
Heart Rate	4. No Adjustment + 7 Genetic PCs	0.012	0.003	[0.005, 0.019]	0.00038
Heart Rate	5. Lead Measures + 7 Genetic PCs	0.011	0.003	[0.005, 0.018]	0.00098
Systolic BP	1. No Adjustment	0.048	0.010	[0.028, 0.068]	3.8e-06
Systolic BP	2. Lead Measures	0.047	0.010	[0.027, 0.068]	4.7e-06
Systolic BP	4. No Adjustment + 7 Genetic PCs	0.009	0.003	[0.002, 0.016]	0.00720
Systolic BP	5. Lead Measures + 7 Genetic PCs	0.009	0.003	[0.002, 0.015]	0.01097
Blood Chemistry and Electrolytes					
HDL Cholesterol	1. No Adjustment	0.274	0.034	[0.208, 0.340]	3.9e-16
HDL Cholesterol	2. Lead Measures	0.277	0.034	[0.209, 0.344]	1.0e-15
HDL Cholesterol	4. No Adjustment + 7 Genetic PCs	0.065	0.007	[0.050, 0.079]	5.2e-18
HDL Cholesterol	5. Lead Measures + 7 Genetic PCs	0.065	0.008	[0.050, 0.080]	1.6e-17
Glucose	1. No Adjustment	0.122	0.020	[0.082, 0.161]	1.9e-09
Glucose	2. Lead Measures	0.120	0.021	[0.080, 0.161]	4.7e-09
Glucose	4. No Adjustment + 7 Genetic PCs	0.013	0.006	[0.002, 0.023]	0.02503
Glucose	5. Lead Measures + 7 Genetic PCs	0.011	0.006	[0.000, 0.023]	0.04344
Carbon Dioxide	1. No Adjustment	0.103	0.021	[0.062, 0.144]	6.9e-07
Carbon Dioxide	2. Lead Measures	0.102	0.021	[0.061, 0.143]	1.1e-06
Carbon Dioxide	4. No Adjustment + 7 Genetic PCs	0.004	0.007	[-0.009, 0.017]	0.52463
Carbon Dioxide	5. Lead Measures + 7 Genetic PCs	0.004	0.007	[-0.009, 0.017]	0.52583
Sodium	1. No Adjustment	0.085	0.018	[0.051, 0.120]	1.2e-06
Sodium	2. Lead Measures	0.086	0.018	[0.051, 0.122]	1.7e-06
Sodium	4. No Adjustment + 7 Genetic PCs	0.008	0.005	[-0.002, 0.019]	0.13258
Sodium	5. Lead Measures + 7 Genetic PCs	0.007	0.006	[-0.003, 0.018]	0.17746
Protein	1. No Adjustment	0.058	0.015	[0.029, 0.086]	7.4e-05
Protein	2. Lead Measures	0.056	0.014	[0.029, 0.084]	6.2e-05
Protein	4. No Adjustment + 7 Genetic PCs	0.023	0.006	[0.011, 0.034]	0.00011
Protein	5. Lead Measures + 7 Genetic PCs	0.023	0.006	[0.011, 0.035]	0.00011
Chloride	1. No Adjustment	0.054	0.012	[0.030, 0.078]	1.2e-05
Chloride	2. Lead Measures	0.054	0.012	[0.029, 0.078]	1.5e-05
Chloride	4. No Adjustment + 7 Genetic PCs	0.013	0.006	[0.002, 0.024]	0.01724
Chloride	5. Lead Measures + 7 Genetic PCs	0.014	0.006	[0.002, 0.025]	0.01674
Organ Function and Blood Cell Markers					
Urea Nitrogen	1. No Adjustment	0.313	0.040	[0.235, 0.391]	2.5e-15
Urea Nitrogen	2. Lead Measures	0.316	0.040	[0.237, 0.394]	3.4e-15
Urea Nitrogen	4. No Adjustment + 7 Genetic PCs	-0.004	0.006	[-0.015, 0.007]	0.44278
Urea Nitrogen	5. Lead Measures + 7 Genetic PCs	-0.004	0.006	[-0.015, 0.007]	0.43995
Alkaline Phosphatase	1. No Adjustment	0.097	0.017	[0.063, 0.131]	2.3e-08
Alkaline Phosphatase	2. Lead Measures	0.093	0.017	[0.060, 0.127]	4.8e-08
Alkaline Phosphatase	4. No Adjustment + 7 Genetic PCs	0.018	0.006	[0.007, 0.030]	0.00190
Alkaline Phosphatase	5. Lead Measures + 7 Genetic PCs	0.017	0.006	[0.005, 0.029]	0.00568
Creatinine	1. No Adjustment	0.075	0.017	[0.043, 0.108]	6.7e-06
Creatinine	2. Lead Measures	0.077	0.017	[0.044, 0.109]	4.9e-06
Creatinine	4. No Adjustment + 7 Genetic PCs	0.014	0.006	[0.003, 0.025]	0.01429

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Table S5 – continued from previous page

Trait	Model	h^2	SE	95% CI	p
Creatinine	5. Lead Measures + 7 Genetic PCs	0.014	0.006	[0.003, 0.025]	0.01318
Alanine Aminotransferase	1. No Adjustment	0.072	0.018	[0.037, 0.107]	4.7e-05
Alanine Aminotransferase	2. Lead Measures	0.071	0.018	[0.036, 0.106]	5.9e-05
Alanine Aminotransferase	4. No Adjustment + 7 Genetic PCs	0.004	0.006	[-0.008, 0.017]	0.48976
Alanine Aminotransferase	5. Lead Measures + 7 Genetic PCs	0.005	0.007	[-0.008, 0.018]	0.45400
Aspartate Aminotransferase	1. No Adjustment	0.022	0.011	[0.001, 0.043]	0.04466
Aspartate Aminotransferase	2. Lead Measures	0.022	0.011	[0.001, 0.043]	0.04033
Aspartate Aminotransferase	4. No Adjustment + 7 Genetic PCs	-0.005	0.006	[-0.017, 0.007]	0.39557
Aspartate Aminotransferase	5. Lead Measures + 7 Genetic PCs	-0.004	0.006	[-0.016, 0.008]	0.47921
Hematocrit	1. No Adjustment	0.020	0.011	[-0.001, 0.041]	0.06152
Hematocrit	2. Lead Measures	0.020	0.011	[-0.002, 0.041]	0.07007
Hematocrit	4. No Adjustment + 7 Genetic PCs	0.004	0.006	[-0.008, 0.016]	0.53741
Hematocrit	5. Lead Measures + 7 Genetic PCs	0.003	0.006	[-0.009, 0.016]	0.61479

Table S6: Haseman–Elston heritability estimates across models with SFOH MCA adjustment.

Trait	Model	h^2	SE	95% CI	p
Anthropometric and Physiological Measures					
Waist Circumference	1. No Adjustment	0.339	0.025	[0.289, 0.389]	8.5e-41
Waist Circumference	2. Lead Measures	0.232	0.021	[0.192, 0.272]	2.5e-29
Waist Circumference	3. Comprehensive Measures	0.188	0.018	[0.152, 0.224]	1.4e-24
Waist Circumference	4. No Adjustment + 7 Genetic PCs	0.033	0.004	[0.025, 0.040]	8.4e-17
Waist Circumference	5. Lead Measures + 7 Genetic PCs	0.029	0.004	[0.021, 0.037]	9.1e-14
Waist Circumference	6. Comprehensive Measures + 7 Genetic PCs	0.028	0.004	[0.020, 0.035]	1.4e-12
Hip Circumference	1. No Adjustment	0.302	0.024	[0.256, 0.349]	7.1e-37
Hip Circumference	2. Lead Measures	0.209	0.020	[0.170, 0.247]	1.7e-26
Hip Circumference	3. Comprehensive Measures	0.165	0.017	[0.132, 0.199]	3.7e-22
Hip Circumference	4. No Adjustment + 7 Genetic PCs	0.031	0.004	[0.023, 0.038]	4.8e-15
Hip Circumference	5. Lead Measures + 7 Genetic PCs	0.028	0.004	[0.020, 0.036]	7.8e-13
Hip Circumference	6. Comprehensive Measures + 7 Genetic PCs	0.027	0.004	[0.019, 0.034]	5.2e-12
Body Mass Index	1. No Adjustment	0.268	0.020	[0.228, 0.308]	3.7e-39
Body Mass Index	2. Lead Measures	0.155	0.015	[0.126, 0.185]	2.3e-25
Body Mass Index	3. Comprehensive Measures	0.112	0.012	[0.088, 0.136]	1.7e-19
Body Mass Index	4. No Adjustment + 7 Genetic PCs	0.036	0.004	[0.029, 0.043]	2.0e-23
Body Mass Index	5. Lead Measures + 7 Genetic PCs	0.032	0.004	[0.025, 0.039]	5.5e-19
Body Mass Index	6. Comprehensive Measures + 7 Genetic PCs	0.029	0.004	[0.022, 0.036]	1.5e-16
Diastolic BP	1. No Adjustment	0.084	0.012	[0.060, 0.109]	1.4e-11
Diastolic BP	2. Lead Measures	0.066	0.011	[0.044, 0.087]	1.5e-09
Diastolic BP	3. Comprehensive Measures	0.057	0.010	[0.038, 0.077]	1.1e-08
Diastolic BP	4. No Adjustment + 7 Genetic PCs	0.012	0.003	[0.006, 0.019]	0.00028
Diastolic BP	5. Lead Measures + 7 Genetic PCs	0.012	0.003	[0.005, 0.018]	0.00056
Diastolic BP	6. Comprehensive Measures + 7 Genetic PCs	0.011	0.003	[0.005, 0.018]	0.00081
Systolic BP	1. No Adjustment	0.048	0.010	[0.028, 0.068]	3.8e-06
Systolic BP	2. Lead Measures	0.037	0.009	[0.019, 0.055]	4.3e-05
Systolic BP	3. Comprehensive Measures	0.031	0.008	[0.014, 0.047]	0.00021
Systolic BP	4. No Adjustment + 7 Genetic PCs	0.009	0.003	[0.002, 0.016]	0.00720
Systolic BP	5. Lead Measures + 7 Genetic PCs	0.009	0.003	[0.002, 0.015]	0.00901
Systolic BP	6. Comprehensive Measures + 7 Genetic PCs	0.008	0.003	[0.002, 0.015]	0.01306
Heart Rate	1. No Adjustment	0.056	0.010	[0.036, 0.076]	2.9e-08
Heart Rate	2. Lead Measures	0.051	0.010	[0.031, 0.071]	3.5e-07
Heart Rate	3. Comprehensive Measures	0.043	0.009	[0.025, 0.061]	1.6e-06
Heart Rate	4. No Adjustment + 7 Genetic PCs	0.012	0.003	[0.005, 0.019]	0.00038
Heart Rate	5. Lead Measures + 7 Genetic PCs	0.011	0.003	[0.005, 0.018]	0.00097
Heart Rate	6. Comprehensive Measures + 7 Genetic PCs	0.011	0.003	[0.004, 0.018]	0.00107
Blood Chemistry and Electrolytes					
HDL Cholesterol	1. No Adjustment	0.274	0.034	[0.208, 0.340]	3.9e-16
HDL Cholesterol	2. Lead Measures	0.178	0.025	[0.128, 0.228]	3.1e-12
HDL Cholesterol	3. Comprehensive Measures	0.135	0.021	[0.094, 0.176]	8.8e-11
HDL Cholesterol	4. No Adjustment + 7 Genetic PCs	0.065	0.007	[0.050, 0.079]	5.2e-18
HDL Cholesterol	5. Lead Measures + 7 Genetic PCs	0.061	0.007	[0.047, 0.076]	2.2e-16
HDL Cholesterol	6. Comprehensive Measures + 7 Genetic PCs	0.059	0.007	[0.045, 0.074]	1.5e-15
Glucose	1. No Adjustment	0.122	0.020	[0.082, 0.161]	1.9e-09
Glucose	2. Lead Measures	0.081	0.016	[0.049, 0.113]	8.7e-07
Glucose	3. Comprehensive Measures	0.069	0.015	[0.039, 0.099]	5.6e-06
Glucose	4. No Adjustment + 7 Genetic PCs	0.013	0.006	[0.002, 0.023]	0.02503
Glucose	5. Lead Measures + 7 Genetic PCs	0.010	0.006	[-0.000, 0.021]	0.06119
Glucose	6. Comprehensive Measures + 7 Genetic PCs	0.009	0.006	[-0.001, 0.020]	0.08829
Carbon Dioxide	1. No Adjustment	0.103	0.021	[0.062, 0.144]	6.9e-07
Carbon Dioxide	2. Lead Measures	0.100	0.021	[0.060, 0.141]	1.0e-06
Carbon Dioxide	3. Comprehensive Measures	0.094	0.020	[0.055, 0.133]	2.4e-06
Carbon Dioxide	4. No Adjustment + 7 Genetic PCs	0.004	0.007	[-0.009, 0.017]	0.52463
Carbon Dioxide	5. Lead Measures + 7 Genetic PCs	0.004	0.007	[-0.009, 0.017]	0.54691
Carbon Dioxide	6. Comprehensive Measures + 7 Genetic PCs	0.003	0.007	[-0.009, 0.016]	0.60079
Sodium	1. No Adjustment	0.085	0.018	[0.051, 0.120]	1.2e-06
Sodium	2. Lead Measures	0.068	0.016	[0.037, 0.098]	1.3e-05
Sodium	3. Comprehensive Measures	0.053	0.014	[0.026, 0.080]	0.00011
Sodium	4. No Adjustment + 7 Genetic PCs	0.008	0.005	[-0.002, 0.019]	0.13258
Sodium	5. Lead Measures + 7 Genetic PCs	0.008	0.005	[-0.003, 0.019]	0.14003

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Table S6 – continued from previous page

Trait	Model	h^2	SE	95% CI	p
Sodium	6. Comprehensive Measures + 7 Genetic PCs	0.007	0.005	[-0.003, 0.018]	0.17269
Protein	1. No Adjustment	0.058	0.015	[0.029, 0.086]	7.4e-05
Protein	2. Lead Measures	0.057	0.014	[0.029, 0.085]	6.6e-05
Protein	3. Comprehensive Measures	0.056	0.014	[0.028, 0.084]	7.5e-05
Protein	4. No Adjustment + 7 Genetic PCs	0.023	0.006	[0.011, 0.034]	0.00011
Protein	5. Lead Measures + 7 Genetic PCs	0.023	0.006	[0.011, 0.034]	0.00012
Protein	6. Comprehensive Measures + 7 Genetic PCs	0.023	0.006	[0.011, 0.034]	0.00012
Chloride	1. No Adjustment	0.054	0.012	[0.030, 0.078]	1.2e-05
Chloride	2. Lead Measures	0.054	0.012	[0.030, 0.078]	1.2e-05
Chloride	3. Comprehensive Measures	0.052	0.012	[0.028, 0.076]	1.6e-05
Chloride	4. No Adjustment + 7 Genetic PCs	0.013	0.006	[0.002, 0.024]	0.01724
Chloride	5. Lead Measures + 7 Genetic PCs	0.014	0.006	[0.003, 0.025]	0.01540
Chloride	6. Comprehensive Measures + 7 Genetic PCs	0.013	0.006	[0.002, 0.024]	0.01787
Organ Function and Blood Cell Markers					
Urea Nitrogen	1. No Adjustment	0.313	0.040	[0.235, 0.391]	2.5e-15
Urea Nitrogen	2. Lead Measures	0.319	0.040	[0.240, 0.397]	1.4e-15
Urea Nitrogen	3. Comprehensive Measures	0.324	0.040	[0.245, 0.403]	8.0e-16
Urea Nitrogen	4. No Adjustment + 7 Genetic PCs	-0.004	0.006	[-0.015, 0.007]	0.44278
Urea Nitrogen	5. Lead Measures + 7 Genetic PCs	-0.004	0.006	[-0.015, 0.006]	0.42682
Urea Nitrogen	6. Comprehensive Measures + 7 Genetic PCs	-0.005	0.006	[-0.015, 0.006]	0.39627
Alkaline Phosphatase	1. No Adjustment	0.097	0.017	[0.063, 0.131]	2.3e-08
Alkaline Phosphatase	2. Lead Measures	0.074	0.014	[0.046, 0.103]	2.6e-07
Alkaline Phosphatase	3. Comprehensive Measures	0.068	0.014	[0.041, 0.095]	7.0e-07
Alkaline Phosphatase	4. No Adjustment + 7 Genetic PCs	0.018	0.006	[0.007, 0.030]	0.00190
Alkaline Phosphatase	5. Lead Measures + 7 Genetic PCs	0.017	0.006	[0.006, 0.029]	0.00339
Alkaline Phosphatase	6. Comprehensive Measures + 7 Genetic PCs	0.017	0.006	[0.006, 0.029]	0.00374
Creatinine	1. No Adjustment	0.075	0.017	[0.043, 0.108]	6.7e-06
Creatinine	2. Lead Measures	0.077	0.017	[0.044, 0.110]	4.9e-06
Creatinine	3. Comprehensive Measures	0.087	0.018	[0.051, 0.123]	1.7e-06
Creatinine	4. No Adjustment + 7 Genetic PCs	0.014	0.006	[0.003, 0.025]	0.01429
Creatinine	5. Lead Measures + 7 Genetic PCs	0.013	0.006	[0.002, 0.024]	0.01646
Creatinine	6. Comprehensive Measures + 7 Genetic PCs	0.013	0.006	[0.002, 0.024]	0.02050
Alanine Aminotransferase	1. No Adjustment	0.072	0.018	[0.037, 0.107]	4.7e-05
Alanine Aminotransferase	2. Lead Measures	0.054	0.016	[0.024, 0.085]	0.00047
Alanine Aminotransferase	3. Comprehensive Measures	0.044	0.014	[0.016, 0.072]	0.00181
Alanine Aminotransferase	4. No Adjustment + 7 Genetic PCs	0.004	0.006	[-0.008, 0.017]	0.48976
Alanine Aminotransferase	5. Lead Measures + 7 Genetic PCs	0.004	0.006	[-0.009, 0.017]	0.52661
Alanine Aminotransferase	6. Comprehensive Measures + 7 Genetic PCs	0.004	0.006	[-0.009, 0.016]	0.56597
Aspartate Aminotransferase	1. No Adjustment	0.022	0.011	[0.001, 0.043]	0.04466
Aspartate Aminotransferase	2. Lead Measures	0.025	0.011	[0.003, 0.048]	0.02792
Aspartate Aminotransferase	3. Comprehensive Measures	0.029	0.012	[0.006, 0.053]	0.01562
Aspartate Aminotransferase	4. No Adjustment + 7 Genetic PCs	-0.005	0.006	[-0.017, 0.007]	0.39557
Aspartate Aminotransferase	5. Lead Measures + 7 Genetic PCs	-0.005	0.006	[-0.017, 0.007]	0.39930
Aspartate Aminotransferase	6. Comprehensive Measures + 7 Genetic PCs	-0.005	0.006	[-0.017, 0.007]	0.39419
Hematocrit	1. No Adjustment	0.020	0.011	[-0.001, 0.041]	0.06152
Hematocrit	2. Lead Measures	0.022	0.011	[0.000, 0.044]	0.04819
Hematocrit	3. Comprehensive Measures	0.024	0.012	[0.001, 0.047]	0.04001
Hematocrit	4. No Adjustment + 7 Genetic PCs	0.004	0.006	[-0.008, 0.016]	0.53741
Hematocrit	5. Lead Measures + 7 Genetic PCs	0.004	0.006	[-0.009, 0.016]	0.56299
Hematocrit	6. Comprehensive Measures + 7 Genetic PCs	0.004	0.006	[-0.009, 0.016]	0.57168

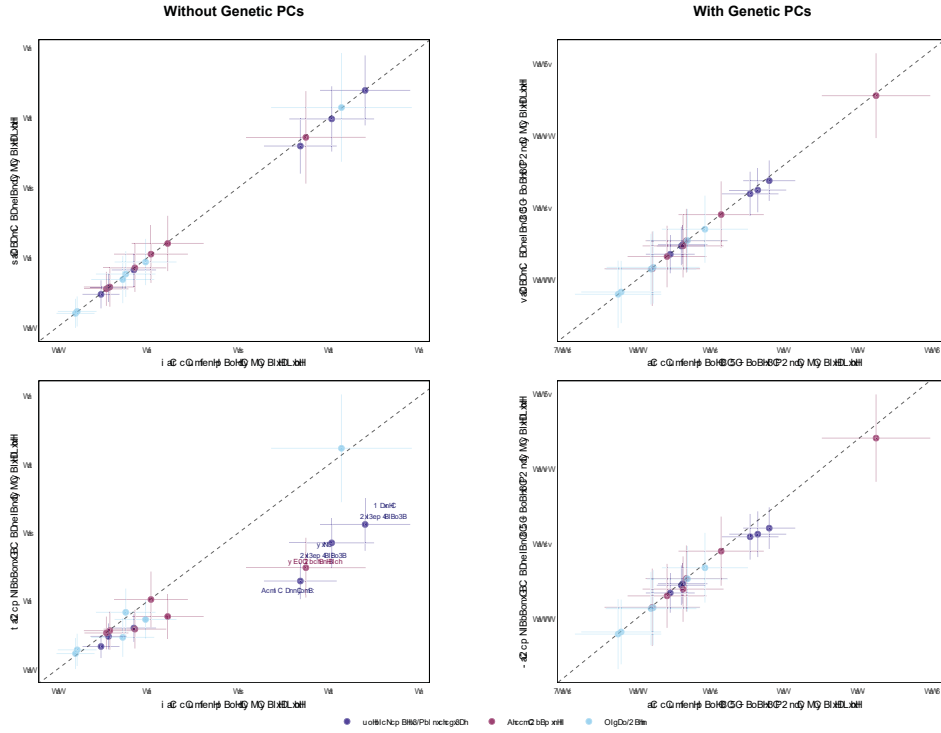


Fig. S5: Combined analysis group Haseman-Elston (HE) regression across SFOH PC adjustment models. HE regression heritability estimates for all relevant participants with 95% confidence intervals for 18 traits, calculated using SFOH PCs. Panels are arranged in a 2×2 grid: models without genetic PCs (left) and with genetic PCs (right). Each panel shows a pairwise model comparison with model numbers noted at the beginning of the x- and y- axis labels. All models are adjusted for age, sex, race, and ethnicity. Models with SFOH adjustment include lead (PCs 1:3) or comprehensive (PCs 1:20) adjustment. Adding SFOH PCs (models 2,3) reduces heritability estimates for BMI, HDL cholesterol, hip circumference, and waist circumference relative to the base model (model 1). Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates the $y = x$ line. Results are broadly consistent with MCA findings, though MCA is the statistically appropriate method for categorical data.

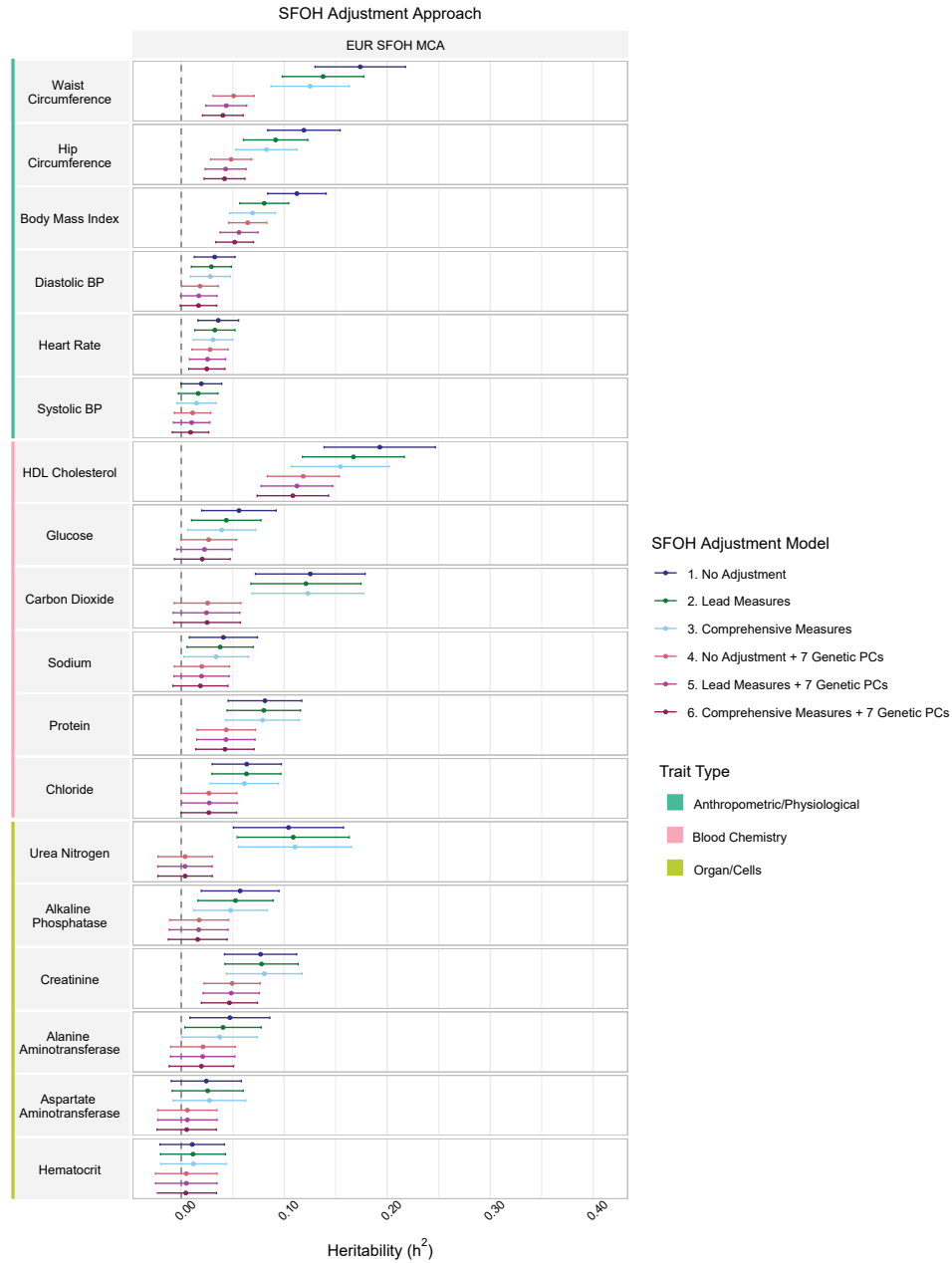


Fig. S6: European-ancestry subgroup Haseman-Elston (HE) regression across SFOH MC adjustment models. HE regression heritability estimates for the European-ancestry subgroup of individuals with 95% confidence intervals for 18 traits (rows), calculated using SFOH MCs calculated from this strict subset of individuals. All models are adjusted for age and sex. Models with SFOH adjustment include lead (strict MCs 1:6) or comprehensive (strict MCs 1:40) adjustment. Patterns are consistent with the combined analysis group (Fig. 2). Points represent HE-CP estimates; intervals are 95% CIs from jack-knife standard errors. The dashed line indicates a heritability of zero.

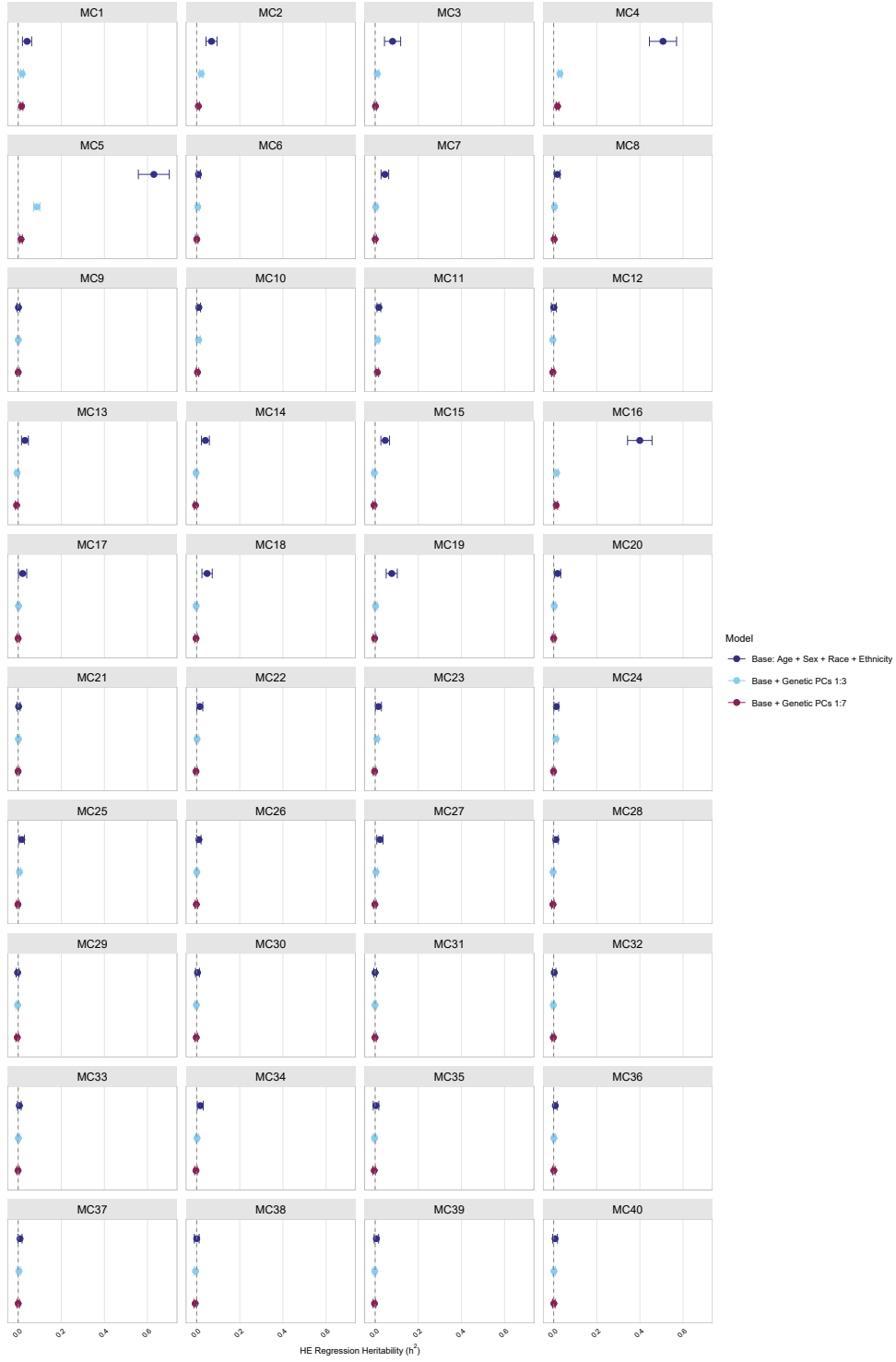


Fig. S7: Combined analysis group Haseman-Elston (HE) regression heritability of SFOH MC axes. HE regression heritability estimates for all relevant participants with 95% confidence intervals for 40 SFOH MC Axes calculated using models with progressive genetic PC adjustment. Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero. HE regression heritability estimates demonstrate dramatic reductions and convergence with genetic PC adjustment indicating these measures capture population stratification.

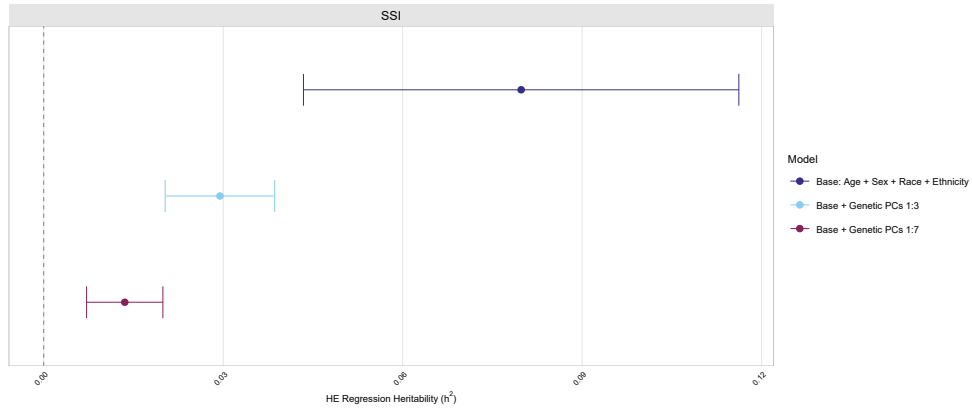


Fig. S8: Combined analysis group Haseman-Elston (HE) regression heritability of the SSI. HE regression heritability estimates for all relevant participants with 95% confidence intervals for the SSI calculated using three nested models: demographic adjustment only, demographics and the top three genetic PCs, and demographics and the top seven genetic PCs. Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero. HE regression heritability estimates show progressive reduction with genetic PC adjustment. While significant at all adjustment levels, the SSI heritability decreases from $h^2 = 0.080$ (base) to 0.029 (3 PCs) to 0.014 (7 PCs), demonstrating that apparent heritability primarily reflects population stratification.

Table S7: HE heritability estimates and p-values across different genetic PC adjustment models. Convergence of SFOH heritability estimates with 7 genetic PCs. Despite appearing to capture different social dimensions, all significantly heritable measures converge to $h^2 \approx 0.01 - 0.02$ after comprehensive genetic PC adjustment, indicating they capture the same underlying population structure rather than distinct genetic influences on different social factors.

SFOH Measure	Base + 7 GenPCs	SE
MC1	0.016	0.003
MC2	0.008	0.003
MC4	0.018	0.003
MC5	0.013	0.003
MC11	0.010	0.003
MC16	0.012	0.003
SSI	0.014	0.003

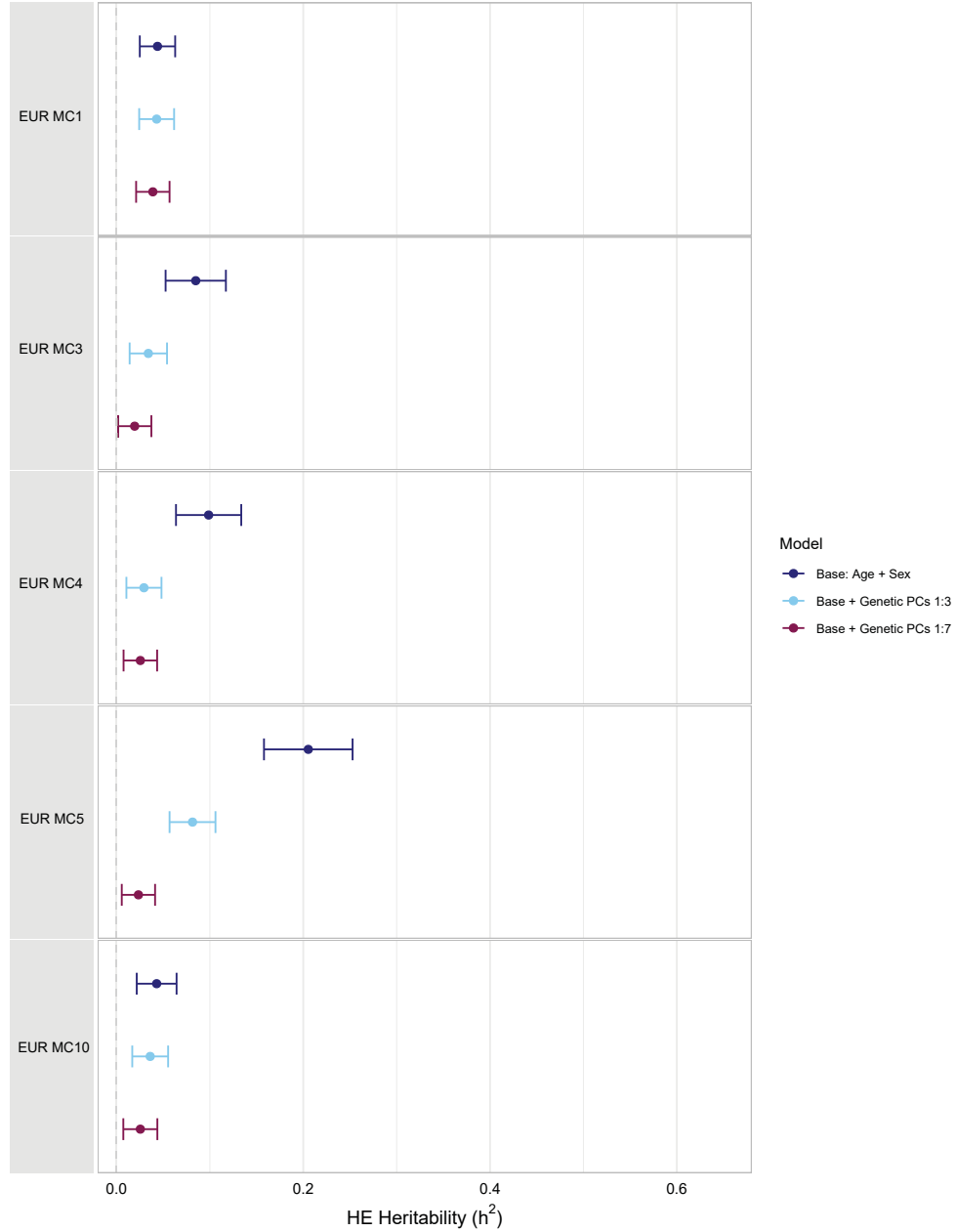


Fig. S9: European-ancestry subgroup Haseman-Elston (HE) regression heritability of SFOH MCs measures calculated in this strict subset that remain heritable after including seven genetic PCs. HE regression heritability estimates for the European-ancestry subgroup of individuals with 95% confidence intervals for SFOH measures that appear heritable after comprehensive genetic PC adjustment. Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero. HE regression heritability estimates show progressive reduction with genetic PC adjustment, demonstrating that apparent heritability primarily reflects population stratification.

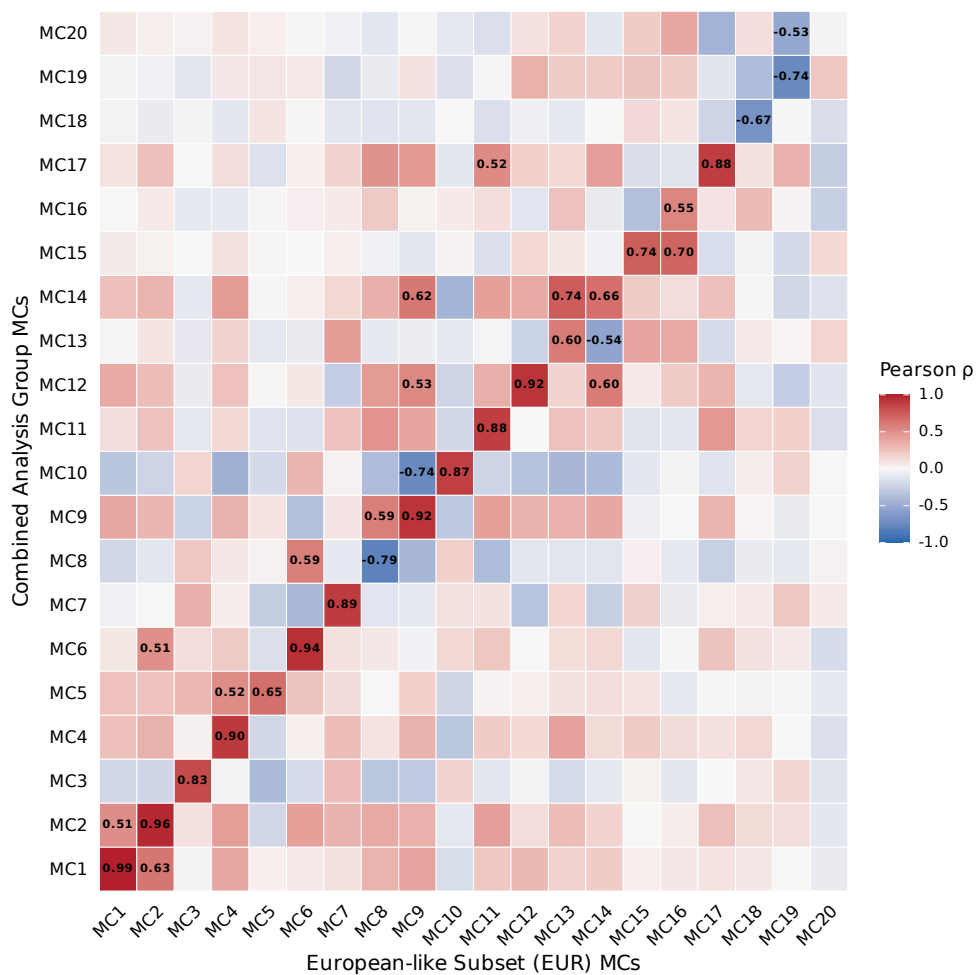


Fig. S10: Heatmap of Pearson correlations between SFOH MCA dimensions from two population subsets. Columns represent the first 20 SFOH MCs derived from the combined analysis group; rows represent corresponding dimensions from the European-ancestry subgroup. The strong correlations on the diagonal indicate that primary latent structures are conserved. Deviations from the diagonal suggest some population-specific variation in some dimensions.

Table S8: Progressive reduction in SFOH measure HE heritability with genetic PC adjustment. Heritability estimates (with respective SEs and p-values) show dramatic reductions and convergence with comprehensive PC adjustment, indicating these measures primarily capture population stratification rather than independent genetic architecture.

SFOH Measure	Base Model			Base + 3 Gen PCs			Base + 7 Gen PCs		
	h^2	SE	p	h^2	SE	p	h^2	SE	p
MC1	0.041	0.011	0.00015	0.018	0.004	2.1e-07	0.016	0.003	1.6e-06
MC2	0.069	0.013	1.2e-07	0.021	0.004	4.4e-07	0.008	0.003	0.01125
MC3	0.081	0.019	2.5e-05	0.009	0.004	0.01718	0.001	0.003	0.69523
MC4	0.507	0.032	2.4e-56	0.029	0.004	2.5e-12	0.018	0.003	5.5e-08
MC5	0.629	0.037	3.0e-66	0.086	0.007	3.7e-33	0.013	0.003	5.8e-05
MC6	0.008	0.005	0.06412	0.005	0.003	0.16431	0.000	0.003	0.87700
MC7	0.045	0.009	5.6e-07	0.003	0.003	0.45194	0.000	0.003	0.98932
MC8	0.017	0.007	0.00827	0.004	0.003	0.26939	0.002	0.003	0.51428
MC9	0.001	0.004	0.74140	0.000	0.003	0.88033	-0.000	0.003	0.98986
MC10	0.010	0.004	0.01854	0.009	0.004	0.01459	0.004	0.003	0.26267
MC11	0.017	0.004	0.00013	0.011	0.003	0.00063	0.010	0.003	0.00170
MC12	0.001	0.006	0.89513	-0.004	0.003	0.22235	-0.004	0.003	0.21539
MC13	0.031	0.008	0.00011	-0.005	0.003	0.10274	-0.007	0.003	0.01201
MC14	0.041	0.009	1.4e-05	-0.003	0.003	0.34876	-0.005	0.003	0.08821
MC15	0.047	0.010	5.8e-06	-0.004	0.003	0.17772	-0.005	0.003	0.08501
MC16	0.399	0.029	2.1e-43	0.013	0.004	0.00013	0.012	0.003	0.00066
MC17	0.021	0.010	0.03207	0.002	0.003	0.62917	-0.000	0.003	0.87682
MC18	0.048	0.012	7.2e-05	-0.002	0.003	0.42587	-0.003	0.003	0.33514
MC19	0.077	0.013	4.6e-09	0.002	0.004	0.65802	-0.003	0.003	0.38524
MC20	0.018	0.007	0.01053	0.003	0.003	0.44374	0.000	0.003	0.99823
MC21	0.002	0.004	0.63232	0.001	0.003	0.74061	-0.001	0.003	0.82097
MC22	0.015	0.007	0.03964	0.002	0.003	0.64023	-0.003	0.003	0.26905
MC23	0.015	0.007	0.03125	0.008	0.004	0.04438	-0.003	0.003	0.38966
MC24	0.012	0.006	0.03012	0.011	0.004	0.00425	-0.001	0.003	0.86288
MC25	0.016	0.006	0.00805	0.006	0.003	0.09796	-0.001	0.003	0.68182
MC26	0.010	0.005	0.04462	0.001	0.003	0.81939	-0.002	0.003	0.58805
MC27	0.022	0.007	0.00307	0.004	0.003	0.27209	-0.001	0.003	0.65345
MC28	0.010	0.006	0.09636	-0.003	0.003	0.40189	-0.003	0.003	0.29030
MC29	-0.003	0.003	0.40979	-0.003	0.003	0.39009	-0.004	0.003	0.15784
MC30	0.004	0.005	0.40037	-0.001	0.003	0.64092	-0.002	0.003	0.50496
MC31	-0.000	0.003	0.95524	-0.001	0.003	0.78493	-0.001	0.003	0.75226
MC32	0.002	0.004	0.53911	-0.002	0.003	0.59799	-0.002	0.003	0.56786
MC33	0.005	0.004	0.22050	0.001	0.003	0.78562	-0.001	0.003	0.81493
MC34	0.017	0.007	0.01587	0.002	0.003	0.60649	-0.003	0.003	0.26291
MC35	0.004	0.007	0.55852	-0.003	0.003	0.34189	-0.005	0.003	0.12258
MC36	0.008	0.004	0.07731	0.001	0.003	0.69476	0.001	0.003	0.80562
MC37	0.008	0.005	0.08437	0.003	0.003	0.29597	0.000	0.003	0.98087
MC38	0.000	0.006	0.97102	-0.005	0.003	0.10106	-0.007	0.003	0.01944
MC39	0.006	0.005	0.27919	-0.002	0.003	0.51705	-0.003	0.003	0.28305
MC40	0.006	0.005	0.23812	0.002	0.003	0.60559	0.001	0.003	0.79116
SSI	0.080	0.019	1.7e-05	0.029	0.005	2.8e-10	0.014	0.003	3.1e-05

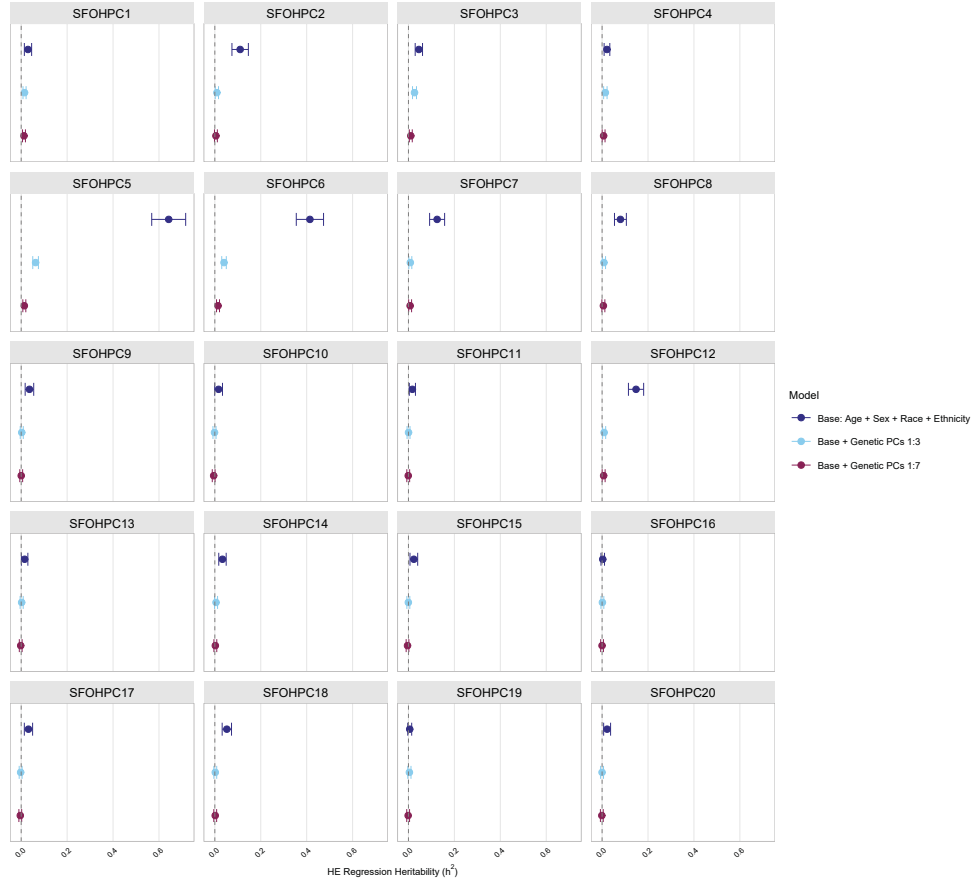


Fig. S11: Combined analysis group Haseman-Elston (HE) regression heritability of SFOH PCs. HE regression heritability estimates for all relevant participants with 95% confidence intervals for 20 SFOH PCs calculated using models with progressive genetic PC adjustment. Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates a heritability of zero. HE regression heritability estimates demonstrate dramatic reductions and convergence with genetic PC adjustment indicating these measures capture population stratification.

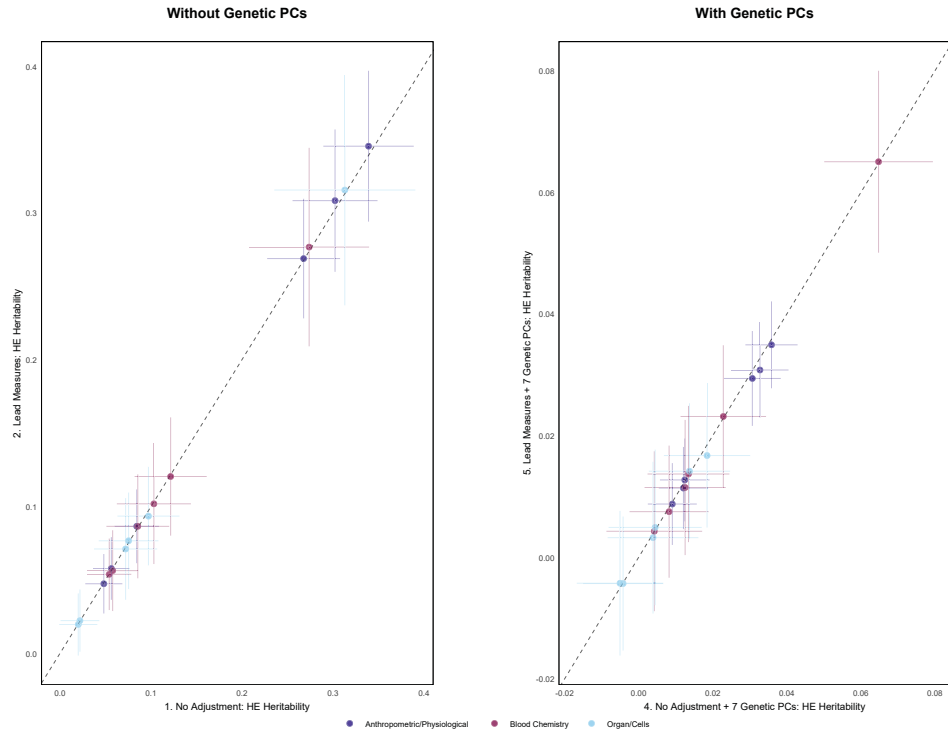


Fig. S12: Combined analysis group Haseman-Elston (HE) regression across SFOH SSI adjustment models. HE regression heritability estimates for all relevant participants with 95% confidence intervals for 18 traits, calculated using SFOH SSI. Models without genetic PCs (left) and with genetic PCs (right). Each panel shows a pairwise model comparison with model numbers noted at the beginning of the x- and y- axis labels. All models are adjusted for age, sex, race, and ethnicity. Adding SFOH SSI (models 2,5) does not reduce heritability estimates relative to the base model (model 1). Points represent HE-CP estimates; intervals are 95% CIs from jackknife standard errors. The dashed line indicates the $y = x$ line.

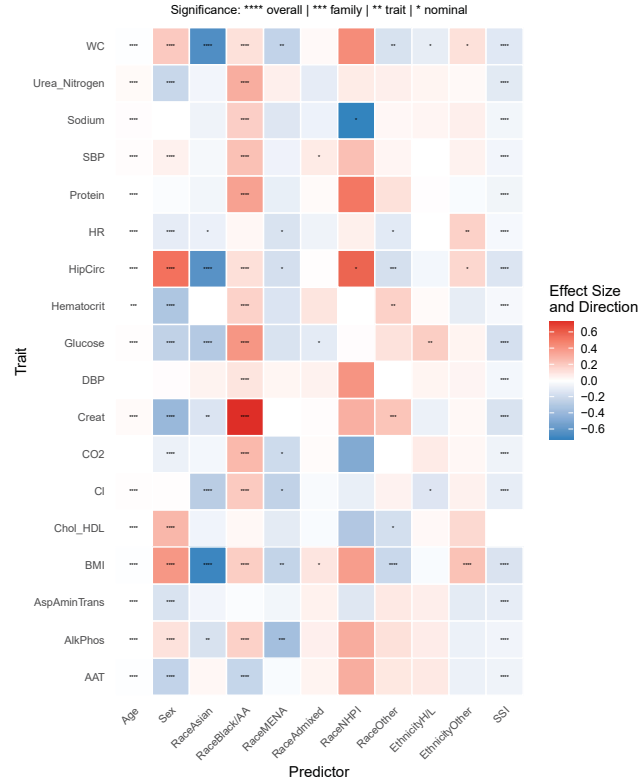


Fig. S13: SFOH SSI dispersion effects. Heatmap of significant predictors in trait (y-axis) dispersion models with demographics and SSI (x-axis). The color gradient (blue to red) and intensity indicates the direction and magnitude of the effect size of the measure on the variance of each trait. Significance is based on a hierarchical Bonferroni correction with four tiers of stringency. Each test was assigned a significance level based on the most stringent threshold it passed and is indicated with “*”.

Table S9: Combined analysis group: Results of regressing SFOH MC axes onto genetic PCs. A linear combination of SFOH MCs explains non-trivial variation in genetic PC1, but limited influence on genetic PCs 2:10.

GENPC	R^2	F-stat	F p-value
GENPC1	0.206	1139.612	0.000
GENPC2	0.039	180.670	0.000
GENPC3	0.043	200.038	0.000
GENPC4	0.031	141.184	0.000
GENPC5	0.011	48.644	0.000
GENPC6	0.017	77.088	0.000
GENPC7	0.004	18.116	0.000
GENPC8	0.008	36.474	0.000
GENPC9	0.004	18.815	0.000
GENPC10	0.003	12.976	0.000

Table S10: Combined analysis group: Results of regressing genetic PCs onto SFOH MC axes. Genetic PCs explain modest amounts of variation in SFOH MC axes structure with the strongest associations observed for SFOH MC3 and MC4.

MC	R^2	F-stat	F p-value
MC1	0.044	408.686	0.000
MC2	0.035	319.490	0.000
MC3	0.076	722.006	0.000
MC4	0.047	434.789	0.000
MC5	0.026	236.145	0.000
MC6	0.008	67.450	0.000
MC7	0.009	81.481	0.000
MC8	0.006	50.074	0.000
MC9	0.003	25.713	0.000
MC10	0.003	28.344	0.000
MC11	0.002	16.505	0.000
MC12	0.001	12.127	0.000
MC13	0.002	16.396	0.000
MC14	0.004	32.517	0.000
MC15	0.003	24.711	0.000
MC16	0.035	319.054	0.000
MC17	0.013	112.996	0.000
MC18	0.016	139.778	0.000
MC19	0.029	258.828	0.000
MC20	0.020	183.289	0.000

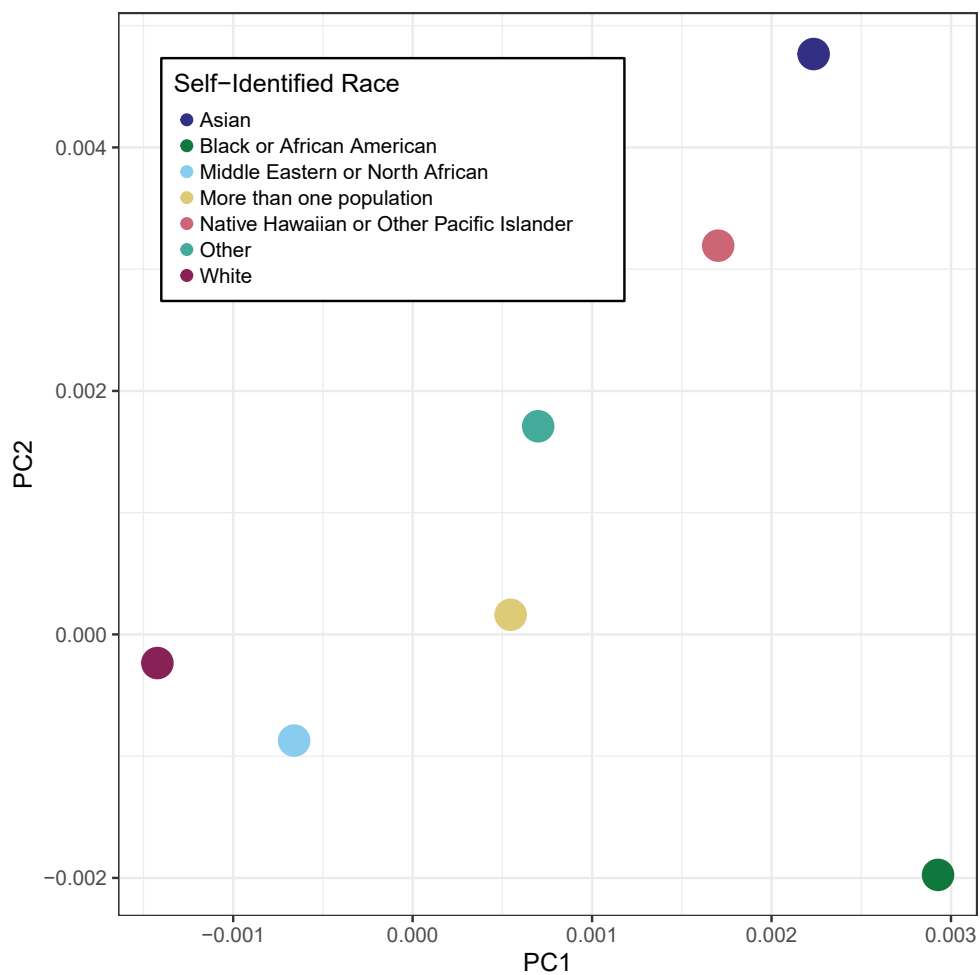


Fig. S14: Population centroids from PCA of the All of Us genetic data. Points represent mean PC1 (x-axis) and PC2 (y-axis) values for each self-identified racial group, summarizing the overall genetic ancestry structure.